Week 2 — Describing data: centre & spread

Business Forecasting

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My code (.R) Notes

Where we are

Describing data → Reasoning under uncertainty → Regression → Time series forecasting

Describing data — centre and spread. Three parts today: first *what kind of data* you have (structure and variable types), then where the data sits (mean, median, mode, percentiles), then how spread out it is — range and IQR, then the measures that use every observation: variance, standard deviation, the coefficient of variation, and the boxplot that puts five of these numbers in one picture.

Part 1 — What structure is the data?

Unit of observation

Every dataset has a unit of observation — *what does one row represent?* Get this wrong and every number after it is wrong.

Ask: *a worker? a firm? a worker-month? a transaction?*

The structure of the data follows from how the unit is observed over time. Three cases — let's see each in a real table.

Three datasets — what is different about them?

Three datasets are loaded: `survey`, `revenue`, `firms`. They record different things in different shapes.

Before we name the shapes: print them and work out what one row is.

Exercise 0 — Print them and decide

Three small data frames are loaded: `survey` (customer satisfaction), `revenue` (one company's accounts) and `firms` (several companies' revenue).

For each one, answer two questions: how many units does it follow, and how many time points does it have per unit?

Exercise | Start Over | Show Hint | Run Code

```
1 survey
2
3 # now the other two:
```

Cross-sectional data

Many units, one point in time — each person responds once, each unit observed only once.

Customer ID	Year	Satisfaction (1-10)	Repurchase?
1	2025	7	Likely
2	2025	8	Unlikely
3	2025	5	Likely
4	2025	9	Likely

Four different customers, all in the same year — the `year` column never changes.

Example: a one-time customer survey. *Business use:* “What drives satisfaction now?”

Time-series (longitudinal) data

One unit, many time points. *“Is it trending?”*

Company	Year	Revenue (MXN)	Employees
Acme	2018	50000	50
Acme	2019	52000	55
Acme	2020	55000	60
Acme	2021	58000	65
Acme	2022	60000	70

One company — `company` never changes — observed in five different years.

Example: one company’s revenue across years. *Business use:* “Forecast next year’s revenue.”

Panel data

Many units, followed over time — cross-section × time-series.

Company	Year	Revenue (MXN)
A	2020	50000
B	2020	60000
C	2020	70000
A	2021	52000
B	2021	63000
C	2021	72000
A	2022	55000
B	2022	65000
C	2022	75000

Three companies and three years — both columns vary, and each company appears once per year.

Example: several companies, each across years. *Business use:* “How did each firm grow?”

Your turn — name the structure

Two more data frames are loaded: `crypto` (market caps of several coins) and `store_visits` (weekly visits to a shop). Print them, then apply the same two counts — how many units, how many time points each.

Exercise | Start Over | Show Hint | Run Code

```
1 # Print them, then count:
```

Part 2 — What kind of variables?

Meet our running dataset: the workforce

We'll use one business dataset all week — a sample of the Mexican labor market, 3,000 workers in 2022, loaded as `workforce`.

You're a compensation analyst: *is pay fair, and how does it vary?*

Nobody has told you what is in it. Open it yourself.

Exercise 1 — Load and look

`workforce` is loaded — 3,000 Mexican workers surveyed in 2022.

Print the first 10 rows of `workforce`. What are the columns called, how many are there, and what does one row represent?

Exercise | Start Over | Show Hint | Run Code

```
1
```

What you just printed

Worker ID	Monthly income (MXN)	Age	Hours/week	Gender	Region	Age group
1	10000	26	20	Men	Rest of Mexico	18-29
2	30100	58	70	Men	Rest of Mexico	50-65
3	6000	30	49	Men	Rest of Mexico	30-39
4	3870	43	9	Women	Rest of Mexico	40-49
5	7740	48	50	Women	Rest of Mexico	40-49
6	5160	35	51	Men	Rest of Mexico	30-39

Six columns. Before we can summarise any of them, we need to know what kind of variable each one is — because the kind decides the summary and the chart.

The four kinds of variable

Every column you will ever meet is one of these:

Categorical — labels or groups, summarised with counts and shares

- Nominal — no natural order *e.g. social platform: Facebook, TikTok, …*
- Ordinal — ordered, but the distances aren't meaningful *e.g. very unsatisfied < … < very satisfied*

Numerical — quantities you can do arithmetic on, summarised with averages and spread

- Discrete — only certain values *e.g. number of vehicles a household owns*
- Continuous — any value in a range *e.g. the MXN/USD exchange rate*

Ordinal vs discrete: both use numbers, but ordinal *labels* have no arithmetic meaning — discrete *counts* do.

Discrete = a countable set of values

A discrete variable can only land on specific values — nothing in between.

- Number of children in a household
- App downloads from the store
- Coke product sizes: 0.33L, 0.5L, 1L, 2.25L



So — what is each of our variables?

`worker_id` · `monthly_income` · `age` · `hours_worked` · `gender` · `region` · `age_group`

One of these is not a variable you would ever summarise. Which — and why not?

Numerical or categorical? Discrete or continuous, nominal or ordinal?

Don’t answer from the names. Get the evidence out of R first.

Exercise 2 — Classify them

`workforce` is loaded — 3,000 Mexican workers, one row each.

Classify the seven columns `workforce$worker_id`, `workforce$monthly_income`, `workforce$age`, `workforce$hours_worked`, `workforce$gender`, `workforce$region` and `workforce$age_group`. Get the evidence first: for a column with only a few distinct values, print its `table()`; for a column with hundreds of values a table would be useless — describe it with `min()`, `max()` and `quantile()` instead.

Try `table(workforce$worker_id)` too, and see what it tells you.

Exercise | Start Over | Show Hint | Run Code

```
1 # Columns with few distinct values: print their table
2
3 # Columns with many distinct values: min, max and a few quantiles
```

Classify our workforce variables

Variable	Type	Subtype
worker_id	Identifier	— never summarised
monthly_income	Numerical	Continuous
age	Numerical	Discrete
hours_worked	Numerical	Discrete
gender	Categorical	Nominal
region	Categorical	Nominal
age_group	Categorical	Ordinal

The type decides the summary and the chart — get it wrong and your client gets a misleading number.

Part 3 — Visualizing categorical data

How is the workforce split across age groups?

The first summary of any categorical variable is a frequency table: the count n_i in each category, and the relative frequency $p_i = n_i / N$.

Build it before you see it.

Exercise 3 — Build the frequency table

`workforce` is loaded — 3,000 Mexican workers.

Make a frequency table of `workforce$age_group` — counts first, then shares. Then do the same for `workforce$region`.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Frequency tables — the polished version

Age group	Workers	Share
18-29	654	21.8%
30-39	883	29.4%
40-49	815	27.2%
50-65	648	21.6%

Region	Workers	Share
Mexico City	59	2.0%
Rest of Mexico	2941	98.0%

Categories are mutually exclusive and collectively exhaustive, so the shares always sum to 1. Exact — but a manager grasps a picture faster.

The table is exact — the picture is faster

A manager will not read a frequency table. Which picture lets them compare age groups at a glance?

Before we show you one: build it yourself.

Exercise 4 — Build the chart

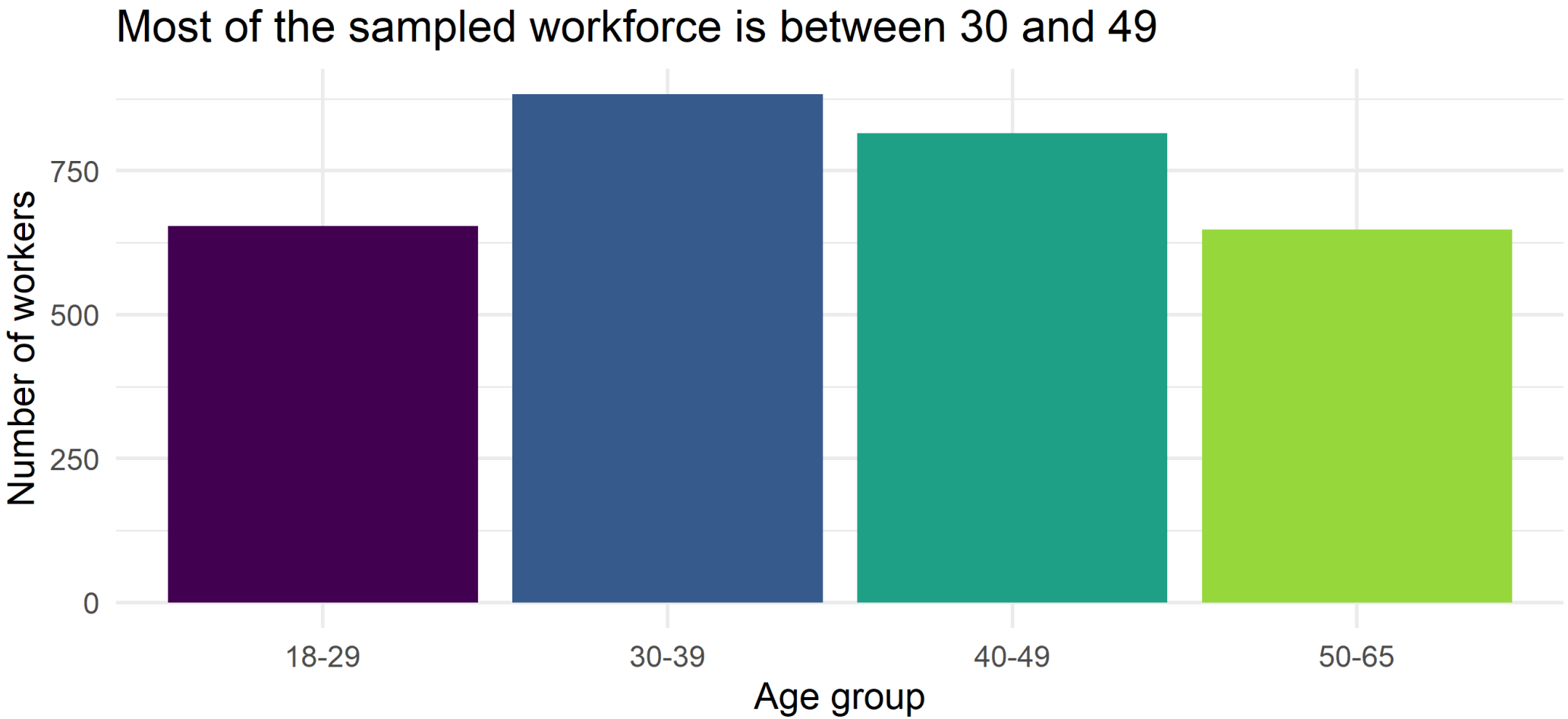
`workforce` is loaded — 3,000 Mexican workers.

Draw the number of workers in each category of `workforce$age_group`. Then do the same for `workforce$region` — does the same recipe work for a variable with no natural order?

Exercise | Start Over | Show Hint | Run Code

```
1  # barplot() takes a TABLE of counts, not the raw column.
```

Bar chart — the polished version



Bars are the right tool for categories: length = count, easy to compare.

Part 4 — Summary statistics: central tendency

Now boil the distribution down to a few numbers managers ask for.

Mean

Mean is the arithmetic average — intuitively, the *balancing point* of the distribution. Hover any underlined piece to see what it means.

Population mean — every one of the N people:

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

Sample mean — the same recipe on the n people you actually observed:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Every observation enters with the same weight $1/n$ — which is exactly why a single huge value drags the mean.

Exercise 7 — The mean

`workforce` is loaded — 3,000 Mexican workers, with monthly pay in `workforce$monthly_income`.

a. Take five incomes typed by hand: 12000, 9500, 15000, 8000, 40000. Apply the formula — add them up and divide by 5 — then check it with `mean()`. Same number. Then compute the mean of `workforce$monthly_income` over the whole sample.

Exercise | Start Over | Show Hint | Run Code

```
1
```

b. Compute the mean of `workforce$gender == "Women"` — what does that number represent?

Exercise | Start Over | Show Solution | Run Code

```
1
```

Median

- The median is the middle value once the data is sorted — half of the observations are below it, half above.
- Odd n: the $\frac{n+1}{2}$-th value. Even n: the average of the two middle values.

Median — seven days of customers

Day	Customers
1	20
2	18
3	25
4	22
5	30
6	21
7	27

$n = 7$ days. What is the median?

- Sort them: 18, 20, 21, 22, 25, 27, 30
- Position: $\frac{7+1}{2} = 4$ → take the 4th value
- 18, 20, 21, 22, 25, 27, 30 → median = 22

Exercise — Change the busiest day

Those seven days again: `c(20, 18, 25, 22, 30, 21, 27)`. Day 5 was the busiest with 30 customers.

Now suppose day 5 had been a huge event: 300 customers instead of 30. Recompute both the mean and the median, and report which one moved.

Exercise | Start Over | Show Hint | Run Code

```
1 customers <- c(20, 18, 25, 22, 30, 21, 27)
2
3 # mean and median as they are:
4
5 # now replace the 30 with 300 and do it again:
```

Exercise 8 — Mean vs median

`workforce` is loaded — 3,000 Mexican workers, monthly pay in `workforce$monthly_income`.

Compute both the mean and the median of `workforce$monthly_income`. That column is right-skewed — which of the two is larger, and why?

Exercise | Start Over | Show Solution | Run Code

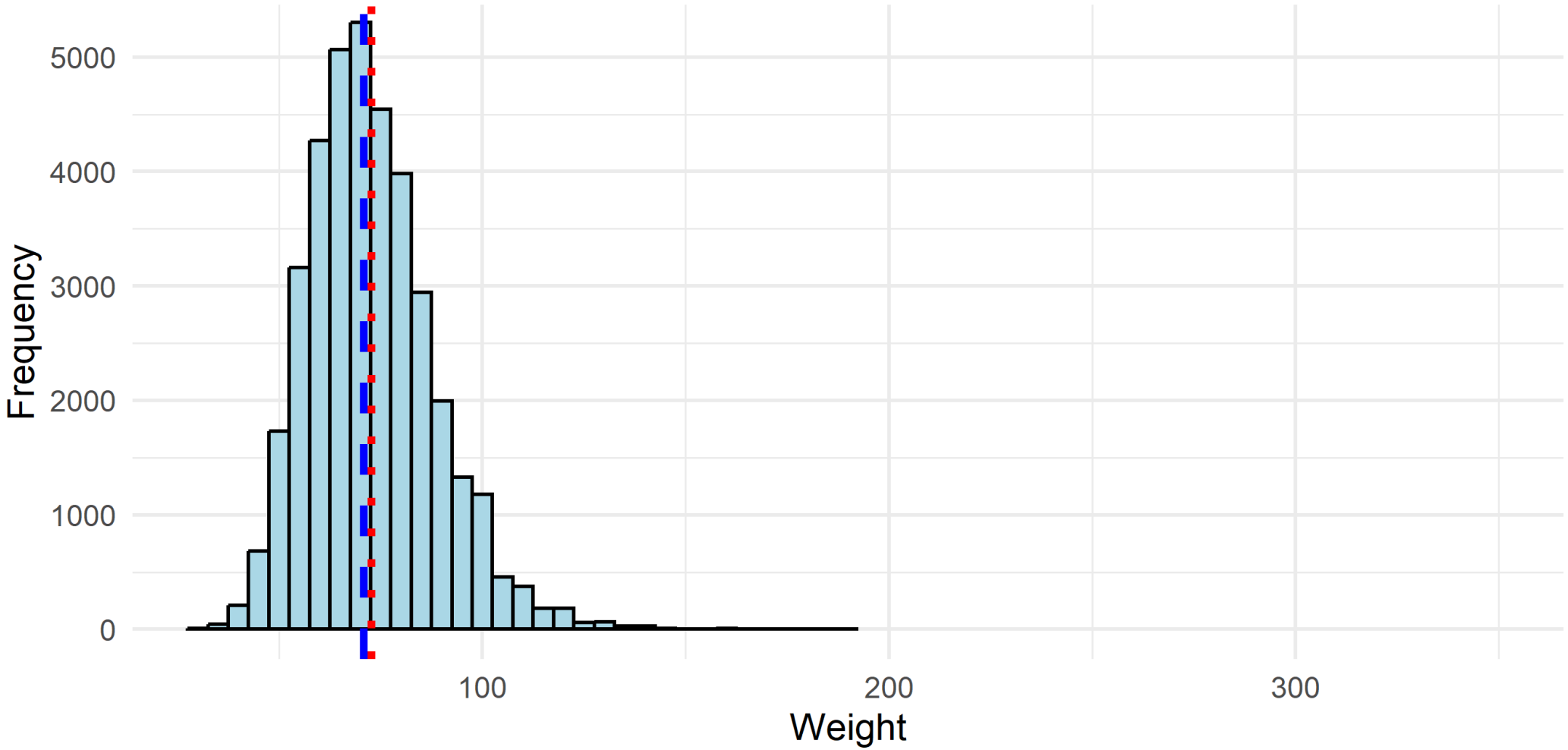
```
1
```

Mean vs Median



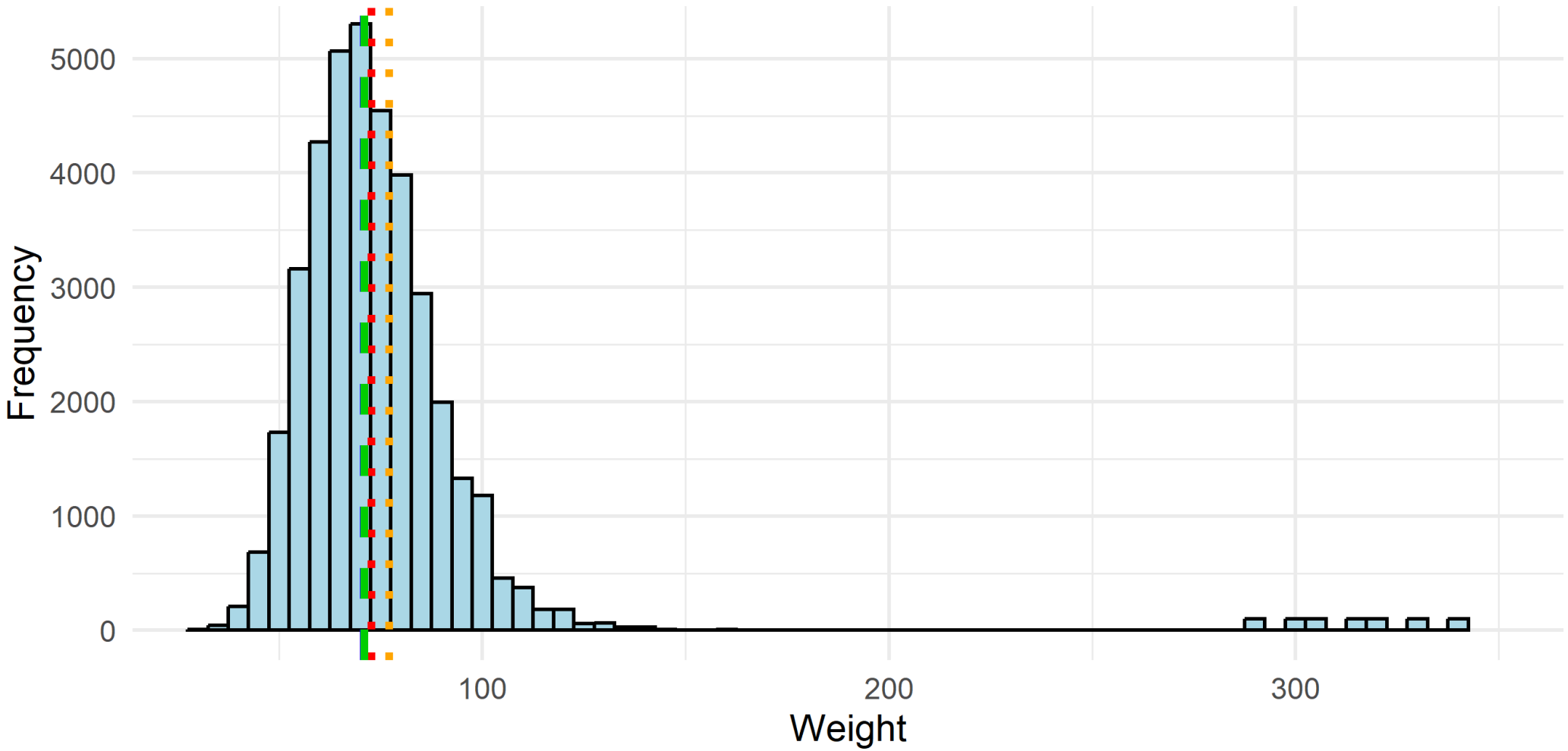
Mean vs median: what an outlier does

Weight in our health data — mean (dotted) and median (dashed):



- Mean: 72.66, Median: 70.75

Now add outliers on the right tail



- Old Mean: 72.66, New Mean: 77.05 — dragged up by the outliers
- Old Median: 70.75, New Median: 70.95 — barely moves

Side note on the Mode

Mode is the most frequent value in the data — the only centre measure that also works for labels, where “average” and “middle” mean nothing.

Exercise 9 — Find the mode

`workforce` is loaded — 3,000 Mexican workers.

Which category of `workforce$age_group` is the most common? And which category of `workforce$region`? Build the frequency table and read the winner off it — name the category out loud, not just the number.

Exercise | Start Over | Show Hint | Run Code

```
1
```

The mode in a frequency table

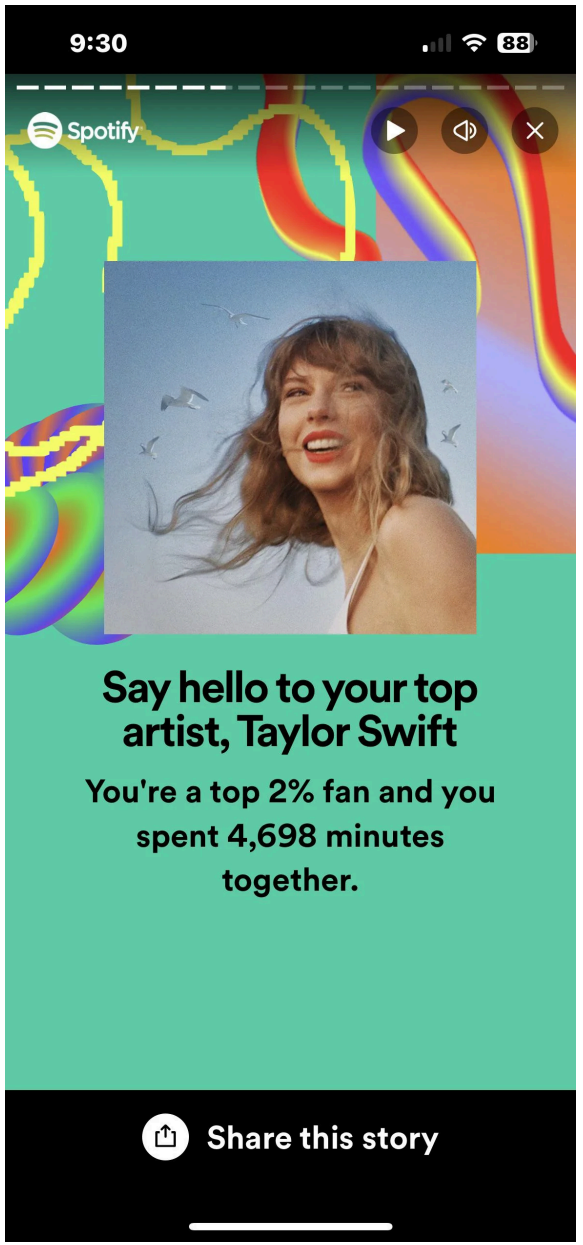
Age of people with diabetes — the modal age is the row with the largest n_i:

Age	n_i	p_i
60	133	0.032
65	129	0.031
56	124	0.030
58	122	0.029
59	119	0.029
68	116	0.028

Percentiles in the wild: a top-2% listener



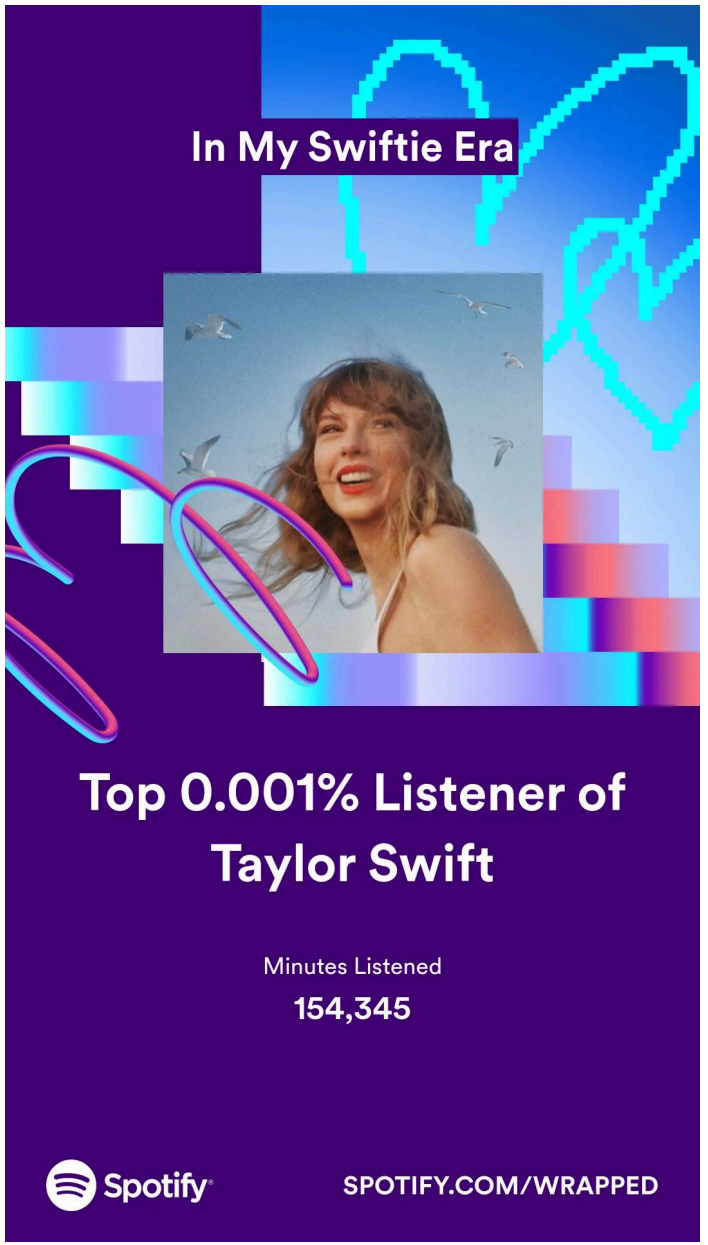
Percentiles in the wild: your top artist



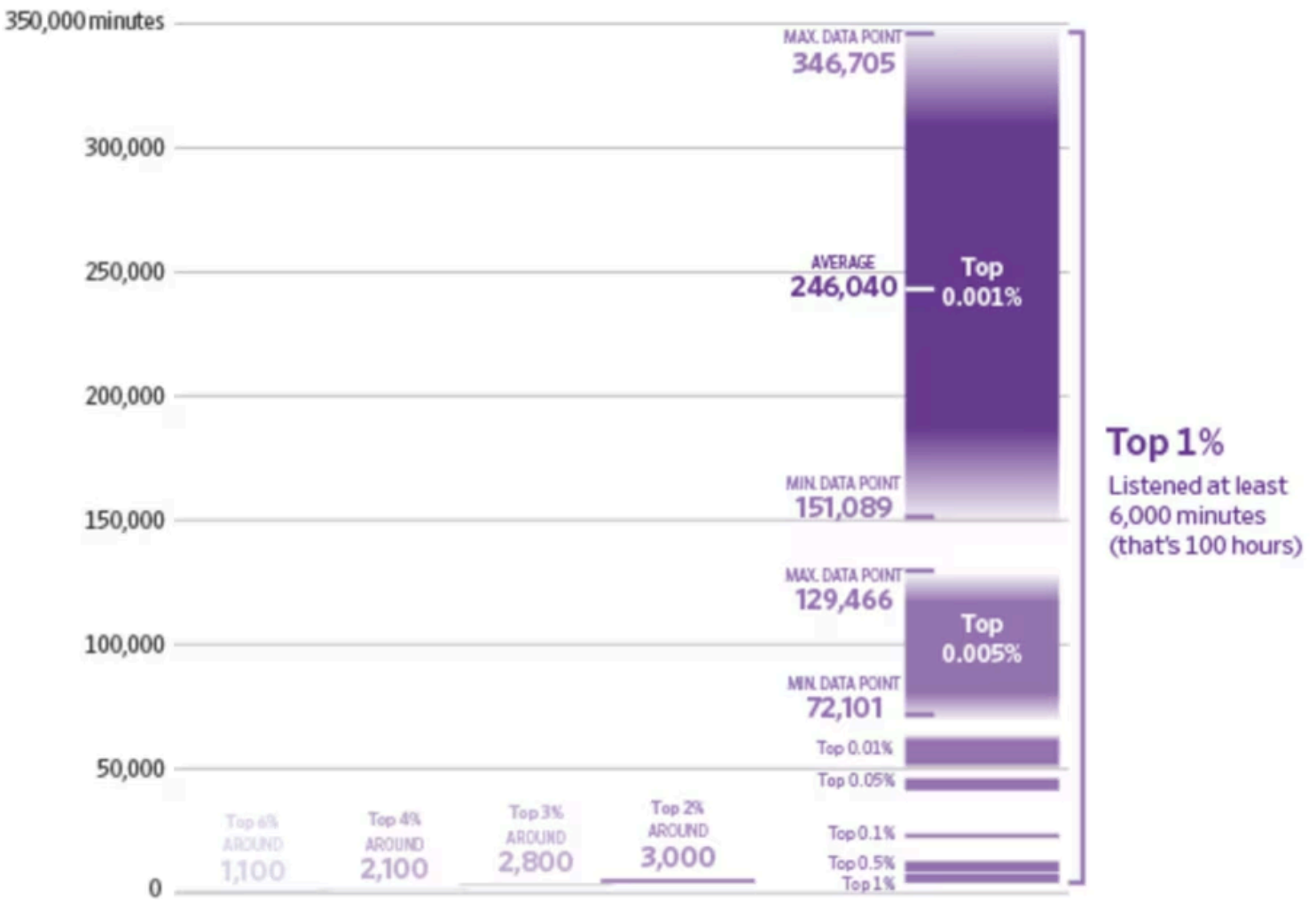
Percentiles in the wild: top 0.1%



Percentiles in the wild: the fine print



Percentiles in the wild: the Wall Street Journal



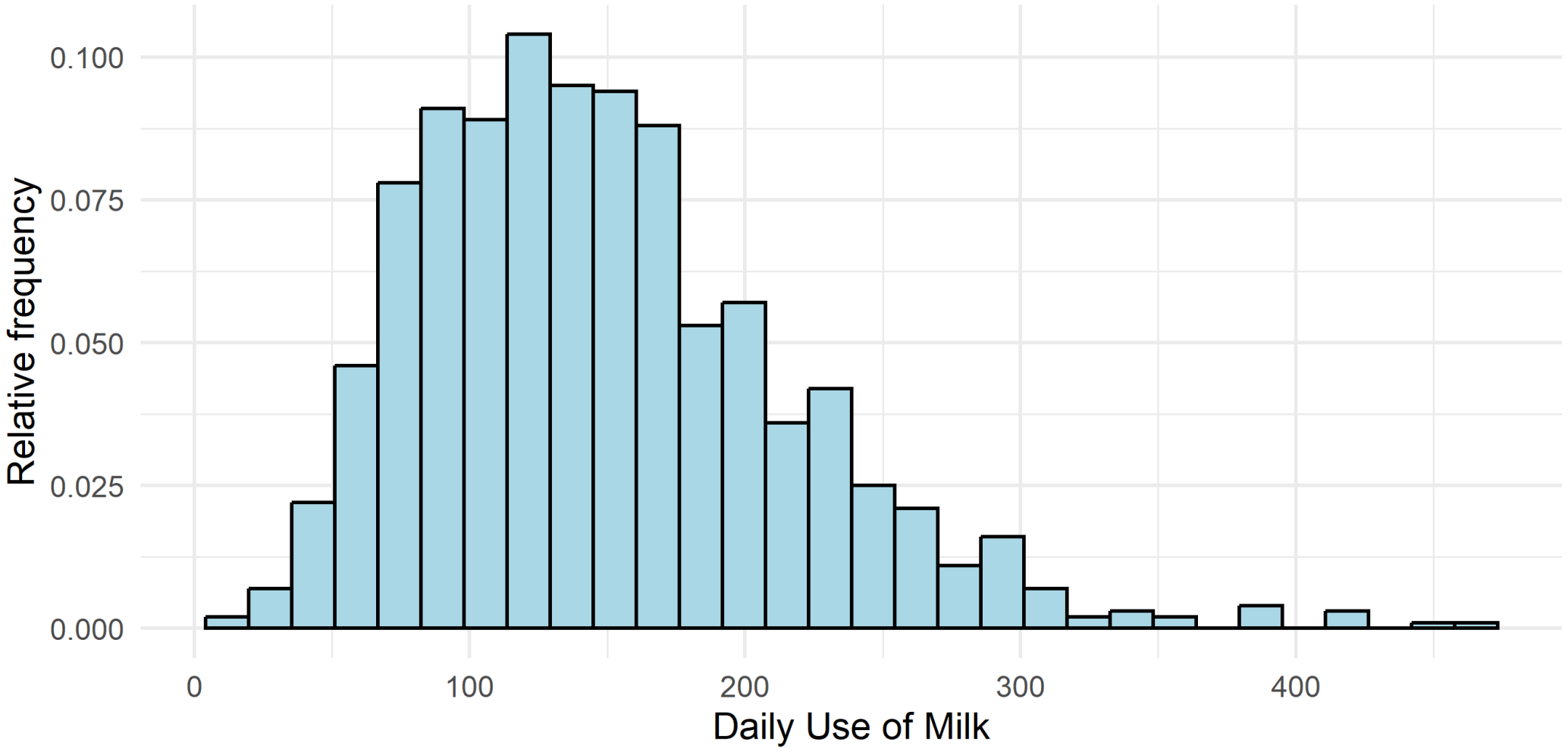
Credits: Wall Street Journal — Article Link

Percentiles at work: how much milk to stock?

- How much inventory of milk you need to keep in your Starbucks?
- What is the tradeoff of keeping too much vs too little inventory?
- Suppose we want to have enough milk to cover sales on 95% of days
- To figure it out, let’s look at the distribution of the daily use of milk

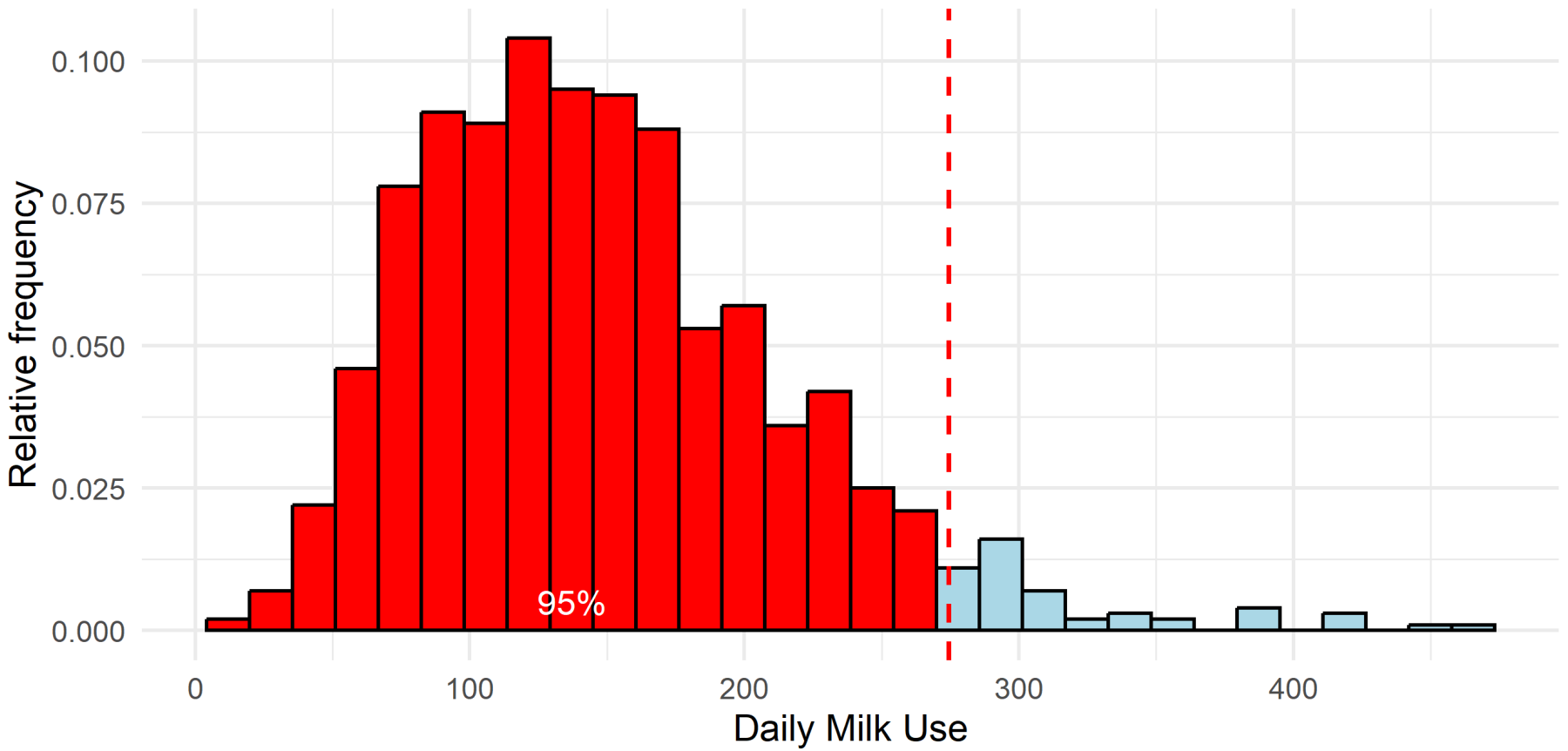
Daily milk use: the distribution

Distribution of the daily use of milk:



Choosing the stock level M

- Let s_i be the daily sales of milk
- We want to choose amount M, such that $P(s_i \leq M) = 0.95$
- That is, in 95% of days sales are smaller or equal than M



The answer is the 95th percentile of the distribution (274 liters).

Percentiles — definition

- *Percentiles* divide the ordered data into 100 equal parts.
- pth percentile is a value such that $p\%$ of the data are below it
  - v_p is such that $P(x_i \leq v_p) = p$
  - v_{95} is such that $P(x_i \leq v_{95}) = 95\%$
- What is the height such that 75% of ITAM students are smaller than this height?
- What is the income level such that 25% of people in Mexico earn less than that level?
- When do I need to leave home to be on time for class 90% of the days?

How to find it in a sample

1. Arrange the data in ascending order

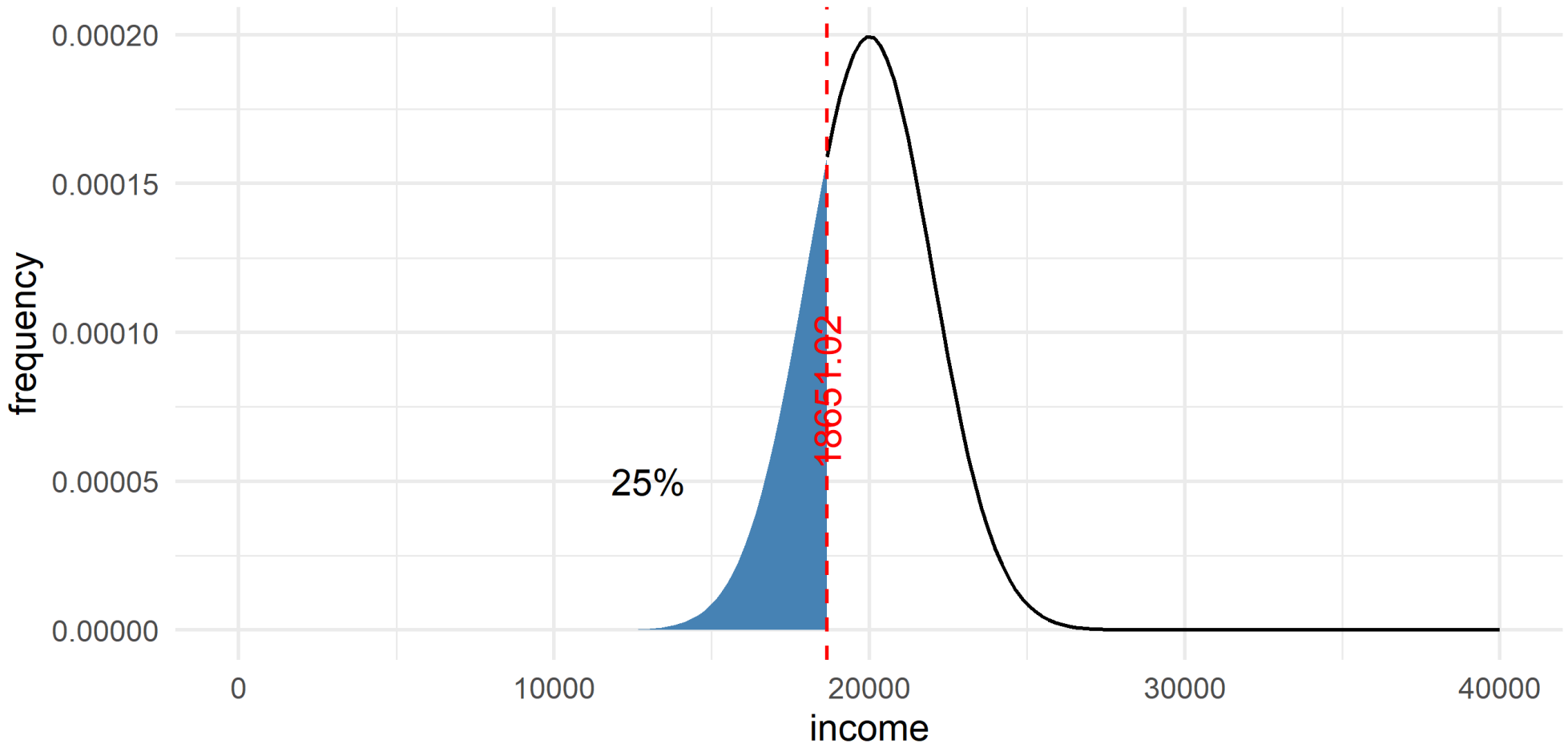
2. Find which observation corresponds to the relevant percentile
   - Formula: $i = \left(\frac{p}{100}\right)(n+1)$
   - Example: 95th percentile in a sample of 1000 observations → $i = \left(\frac{95}{100}\right)(1000+1) = 950.95$

3. If it's an integer, the value of the i-th observation is your percentile
4. If it's not, take the average between i rounded down and i rounded up
   - Here: the average of the 950th and 951st observation

Why $n+1$ and not n? Because n sorted observations cut the line into $n+1$ gaps, so the k-th observation marks the $k/(n+1)$ point of the distribution — and it is the version that puts the median exactly on the middle observation ($n = 7$: $i = 0.5 \times 8 = 4$ ✓).

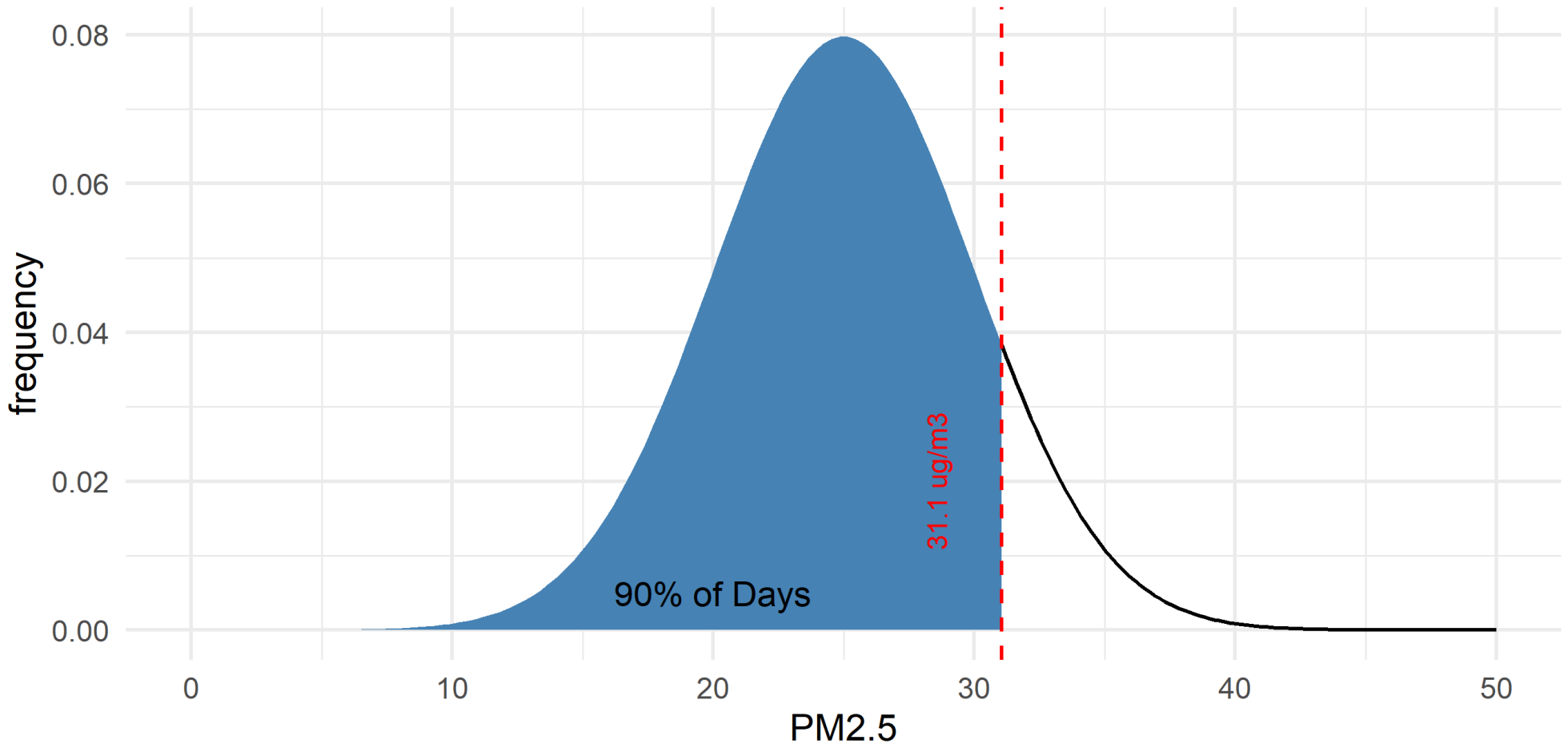
Common values

- Median — 50th percentile — half of the values are below the median
- Quartiles — 25th, 50th and 75th percentile.
  - How poor is the poorest quartile of the society? Their income is below the 25th percentile.



Common values (cont.)

- Deciles — 10th, 20th, … 90th
  - How bad does pollution get in CDMX during the top 10% polluted days?
  - During the top 10% of polluted days, pollution is larger than the 9th decile.



Exercise 10a — Percentile by hand

No data frame here — you type the numbers yourself into a vector called `x`: daily milk use (liters) in a café over 9 days, 120, 85, 190, 140, 95, 160, 210, 130, 105.

Find the 25th percentile of `x` with the formula. Step 1 of the recipe is *arrange the data in ascending order* — do that yourself and type the sorted values into `c(...)`. Then the index is $i = \left(\frac{p}{100}\right)(n+1)$ with $p = 25$ and $n = 9$. It will not come out a whole number, so average the two values of `x` on either side of it.

Exercise | Start Over | Show Hint | Run Code

```
# Step 1: type the nine values in ascending order
x <- c(______)

# Step 2: the index  i = (25/100) * (9 + 1)

# Step 3: i is not a whole number -> average the values at the two positions around it

# Step 4: check yourself against quantile()
```

Exercise 10b — Percentiles on real data

`workforce` is loaded — 3,000 Mexican workers, monthly pay in `workforce$monthly_income`.

Compute the 10th, 25th, 50th, 75th, and 90th percentiles of `workforce$monthly_income` with `quantile()`. The 90th is the income 90% of the 3,000 workers earn below.

Exercise | Start Over | Show Solution | Run Code

```
1
```

Part 5 — First measures of spread

Why dispersion matters — how would you measure it?

- Suppose a store has an average daily revenue of 10,000 pesos
- It could take exactly 10,000 every single day
- Or take 20,000 on half the days and 0 on the other half
- Same average, completely different business. Dispersion is what tells them apart — and it is what risk is made of.

So: how would you put a number on it?

Range

$$R = x_{max} - x_{min}$$

The distance between the two extremes: the simplest measure of spread there is.

Two questions to put it to work:

- How far apart are the lowest-paid and the highest-paid women in `workforce`?
- Is women's pay more or less spread out than men's?

Exercise 11 — Range

`workforce` is loaded — 3,000 Mexican workers, with monthly pay in `workforce$monthly_income`.

How far apart are the lowest-paid and the highest-paid worker? Find both ends, then the distance between them.

Exercise | Start Over | Show Hint | Run Code

```
# the two ends:

# the range:
```

Range: what you found

$$R = 86000 - 133 = 85867 \text{ pesos}$$

A huge number — but look at what produced it: two people. The highest earner and the lowest earner. The other 2,998 workers had no say in it at all.

Change the top earner's pay and the range changes with it. That is the weakness we fix next.

Interquartile Range

- Interquartile range is the difference between the first and the third quartile of the data:

$$IQR = q_3 - q_1$$

- q_1 and q_3 are the 25th and 75th percentiles we just computed with `quantile()`.
- What is the IQR of `workforce$monthly_income` in each group — and does it tell the same story the range told?
- And what share of a group earns between q_1 and q_3?

Exercise 12 — Interquartile range

`workforce` is loaded — 3,000 Mexican workers, with monthly pay in `workforce$monthly_income`.

Get q_1 and q_3 of `workforce$monthly_income` with `quantile()` and take their difference. Then compute the share of workers whose pay lies between those two numbers — write down what it *should* be before you run it.

Exercise | Start Over | Show Hint | Run Code

```
1 # my guess for the share between q1 and q3: ____
```

Interquartile range: what you found

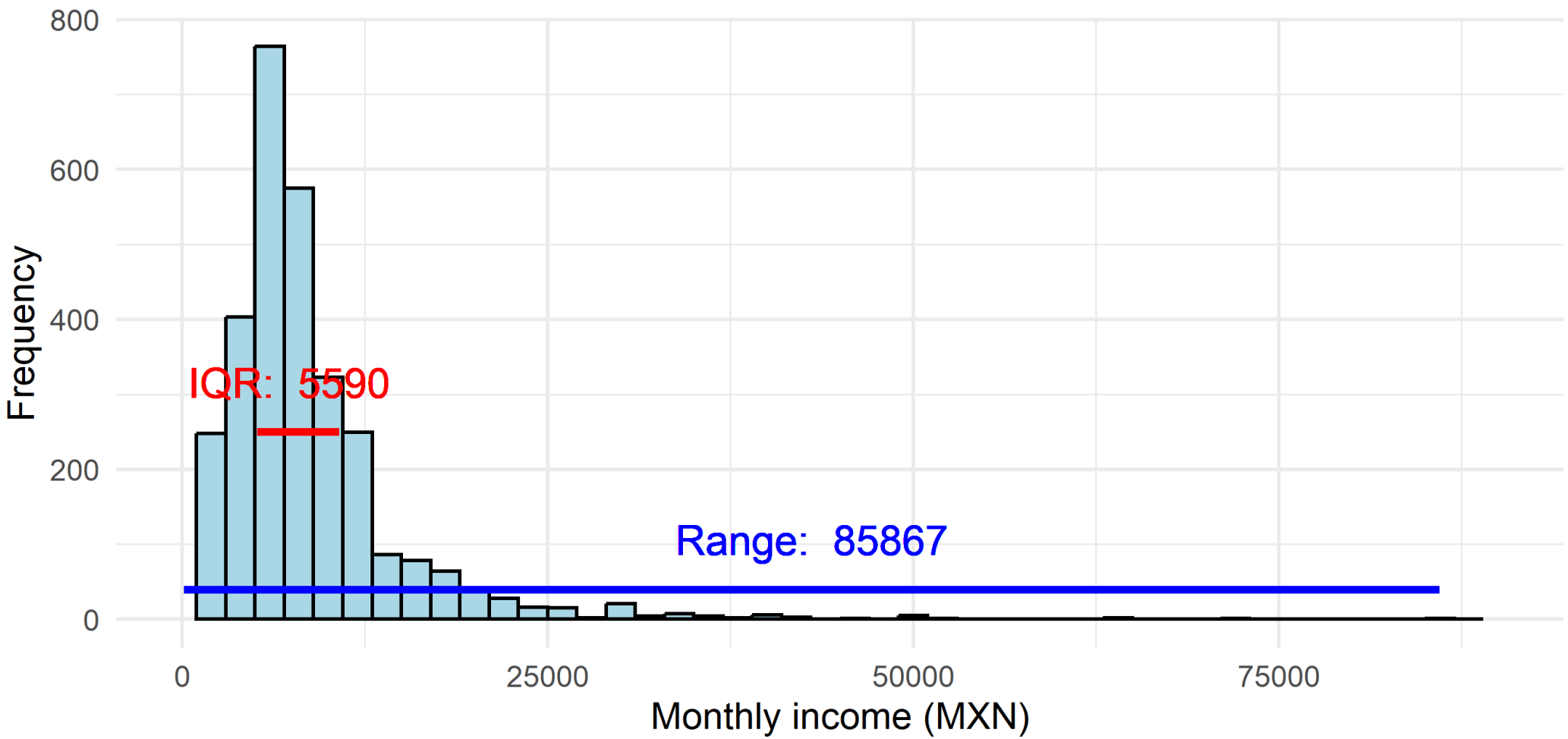
$$IQR = q_3 - q_1 = 10750 - 5160 = 5590 \text{ pesos}$$

Compare that with the range of 85 867 pesos you found a moment ago. Same data, same question, an answer many times smaller — because the IQR describes the middle half instead of the two extremes.

And the share of workers between q_1 and q_3 came out at about half — which is exactly what the quartiles promise.

Range and IQR on the same picture

The blue bar spans everything; the red bar spans the middle half.

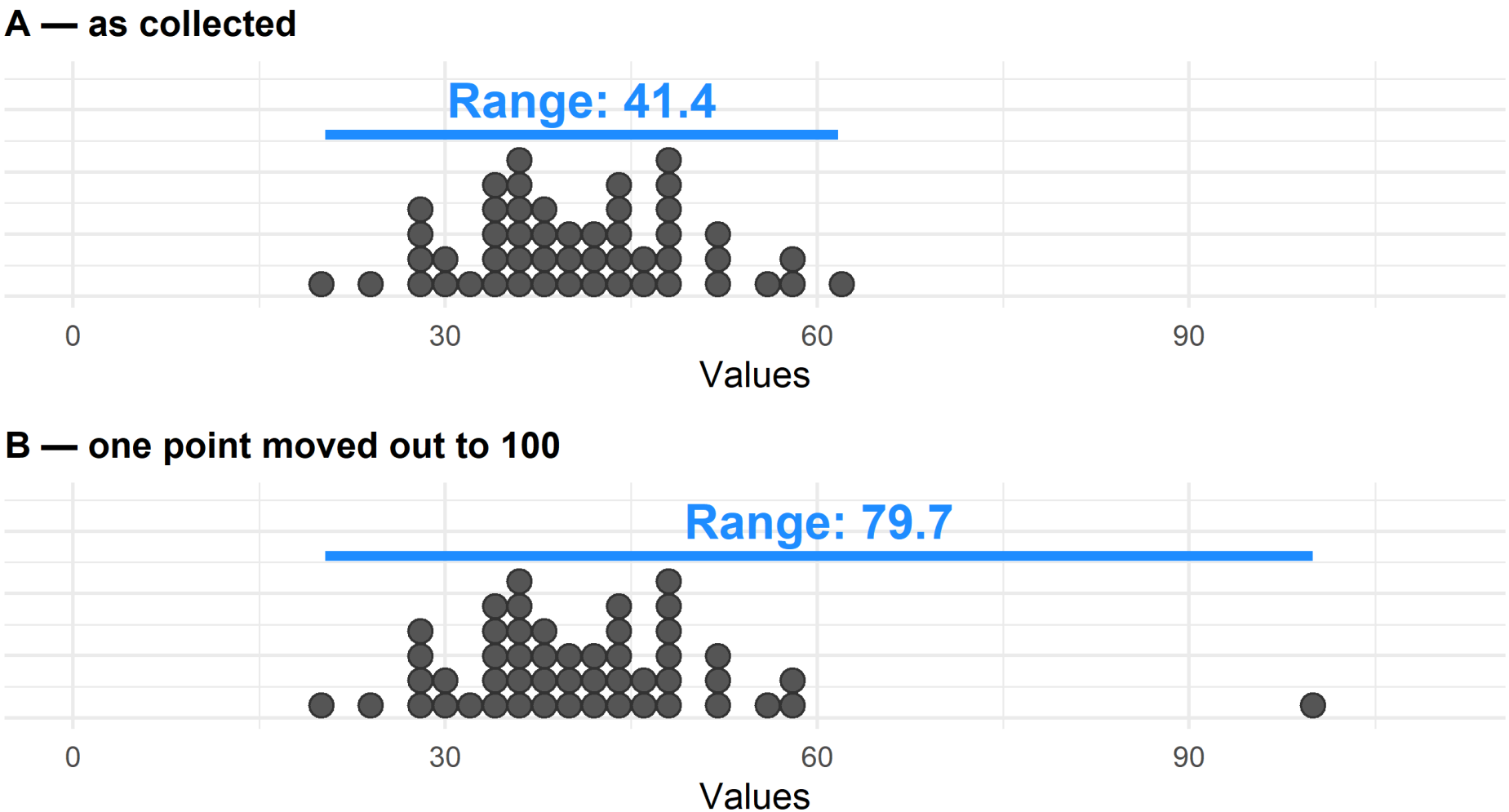


One typo, four statistics

A payroll clerk enters a worker’s monthly income with one extra zero: 100 000 pesos instead of 10 000.

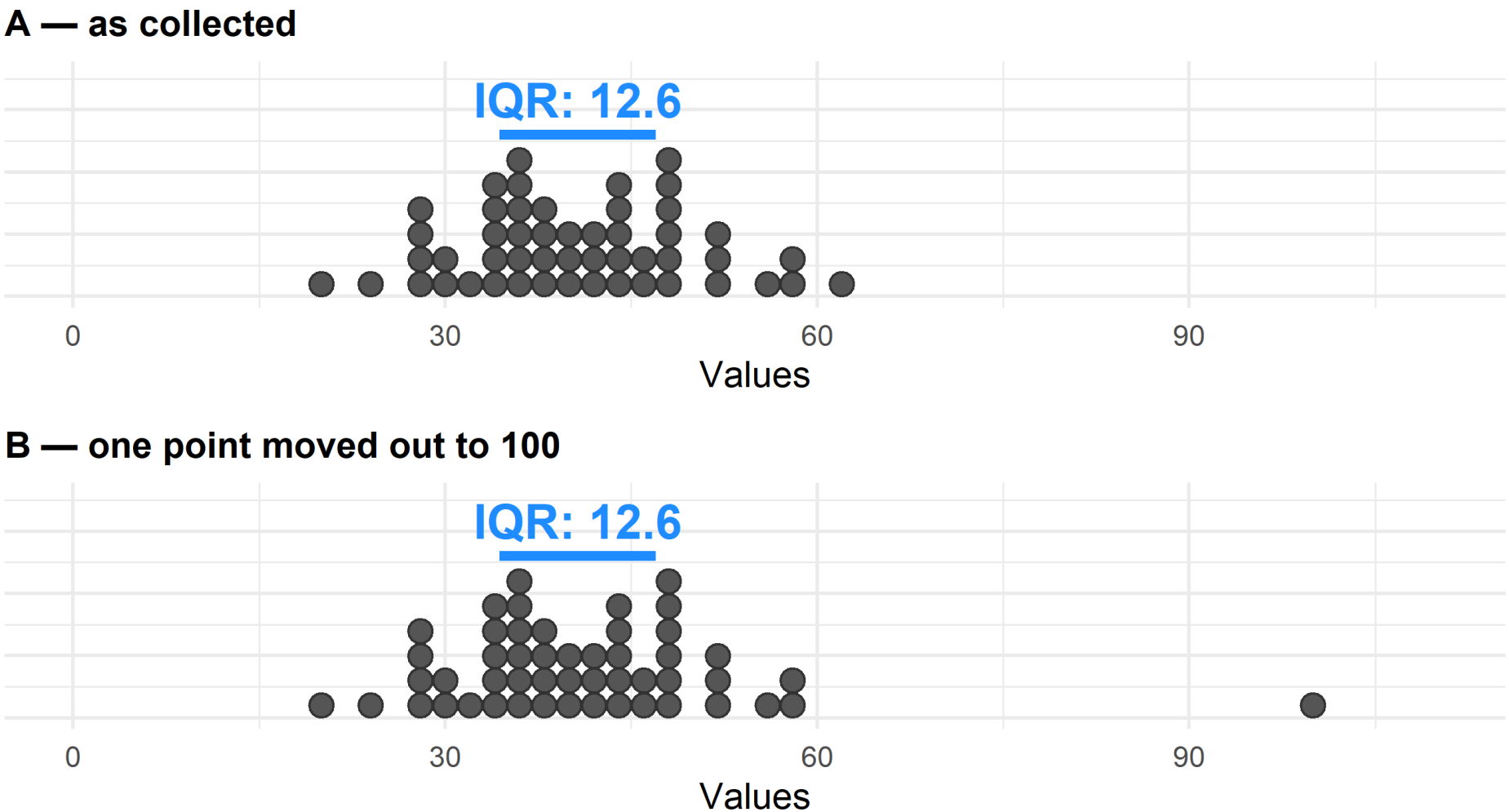
Which of these does that single keystroke move — the mean, the median, the range, the IQR?

Range: sensitive to outliers



- Same 50 observations in both panels — B just moves one point out to 100.
- The range is very sensitive to outliers: it is built from the two extremes.

IQR: robust to outliers



- Same two panels, now with the IQR drawn instead of the range.
- The IQR is robust: it hardly notices the moved point — same number, twice.

Part 6 — Spread that uses every observation

The range and the IQR both threw away most of the data. Here we meet the spread measures that use every observation, and ask how they react when the data is rescaled.

Variance

The range and the IQR each used two numbers. Now we want a measure that uses every observation: how far is a typical observation from the mean?

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

Written in probability language, the same thing is $E[(X-\mu)^2]$ — and for a discrete variable, $\sigma^2 = \sum_k P(X=k)(k-\mu)^2$.

But why squared? Why not simply add up the distances $(x_i - \mu)$ and average them?

Hold that thought — try it on paper for two or three numbers and see what you get.

Sample equivalents

- Sample variance:

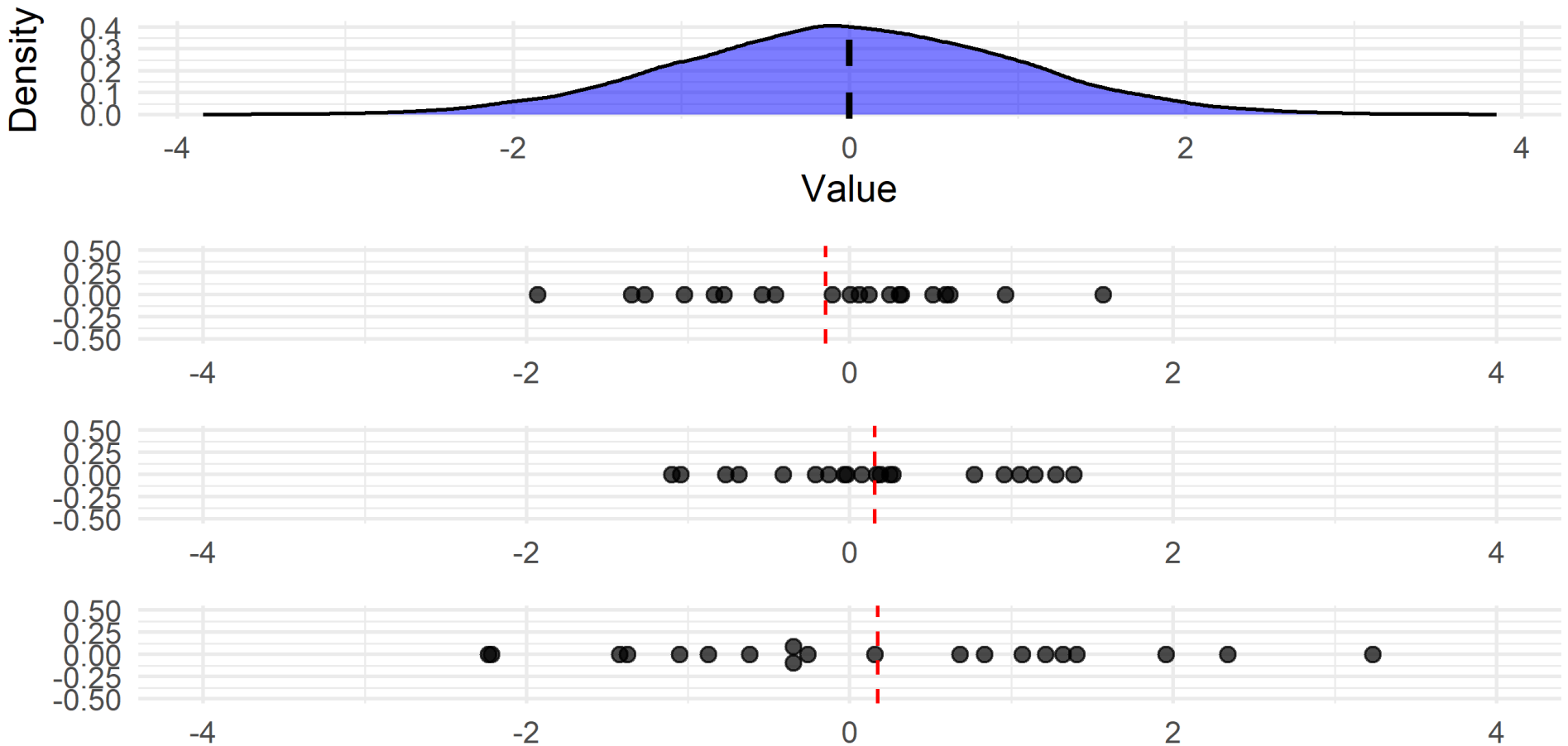
$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2)$$

- **Sample standard deviation**:

$$s = \sqrt{\frac{1}{n-1}(\sum_{i=1}^{n} x_i - \bar{x})^2} = \sqrt{\frac{1}{n-1}(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2)}$$

- Why we divide by $n-1$ rather than n?

Why n−1: small samples

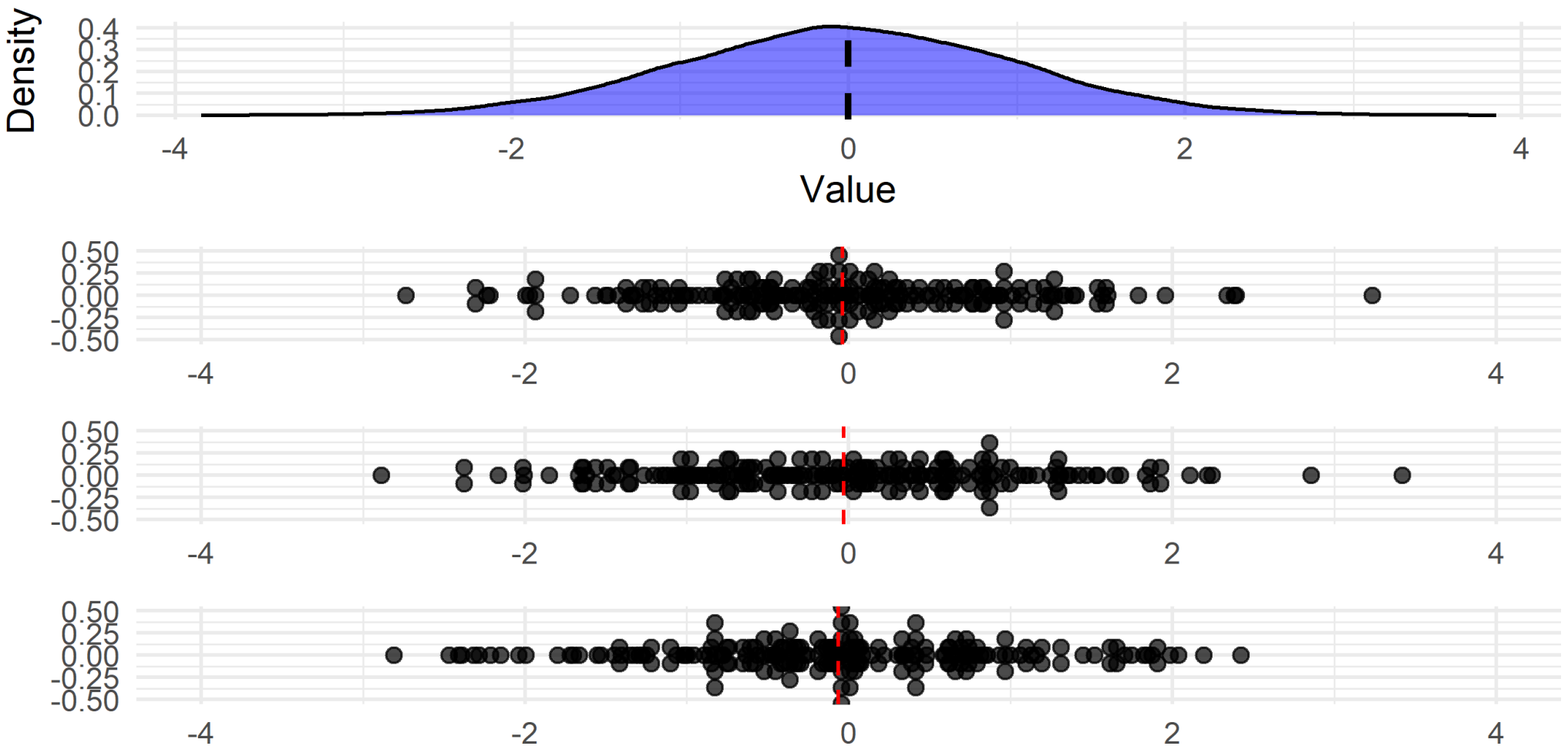


- Top panel: the population, with the population mean (black). Below: three small samples of 20, each with its own sample mean (red).
- Intuition — observed values usually fall closer to their own sample mean than to the population mean. The distances we measure are artificially small.

Sample equivalents (cont.)

- So the deviations from the sample mean underestimate the population standard deviation
- So we divide by a smaller number to correct for it
- In big sample $\frac{1}{n}$ and $\frac{1}{n-1}$ are similar, so correction doesn't matter as much

Why n−1: big samples



- The same picture with samples of 200 instead of 20.
- Intuition — in big samples our estimate of the population mean is already good, so there is little left to correct.

A manager asks: “spread of what, exactly?”

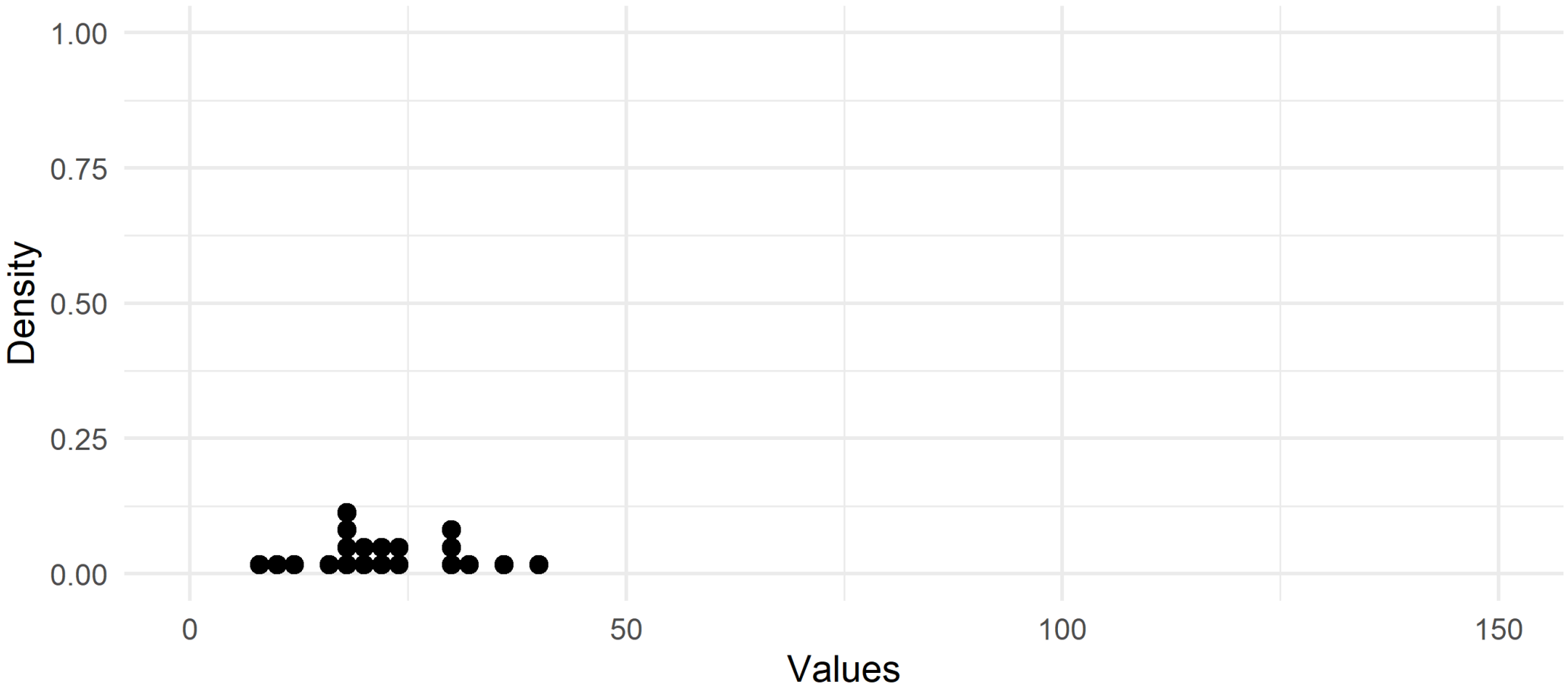
You hand your manager the variance of `Health_data$weight`. She asks: *“and that’s in kilograms, right?”*

Is it? And what would happen to that number if the hospital recorded weights in grams instead?

Standard Deviation

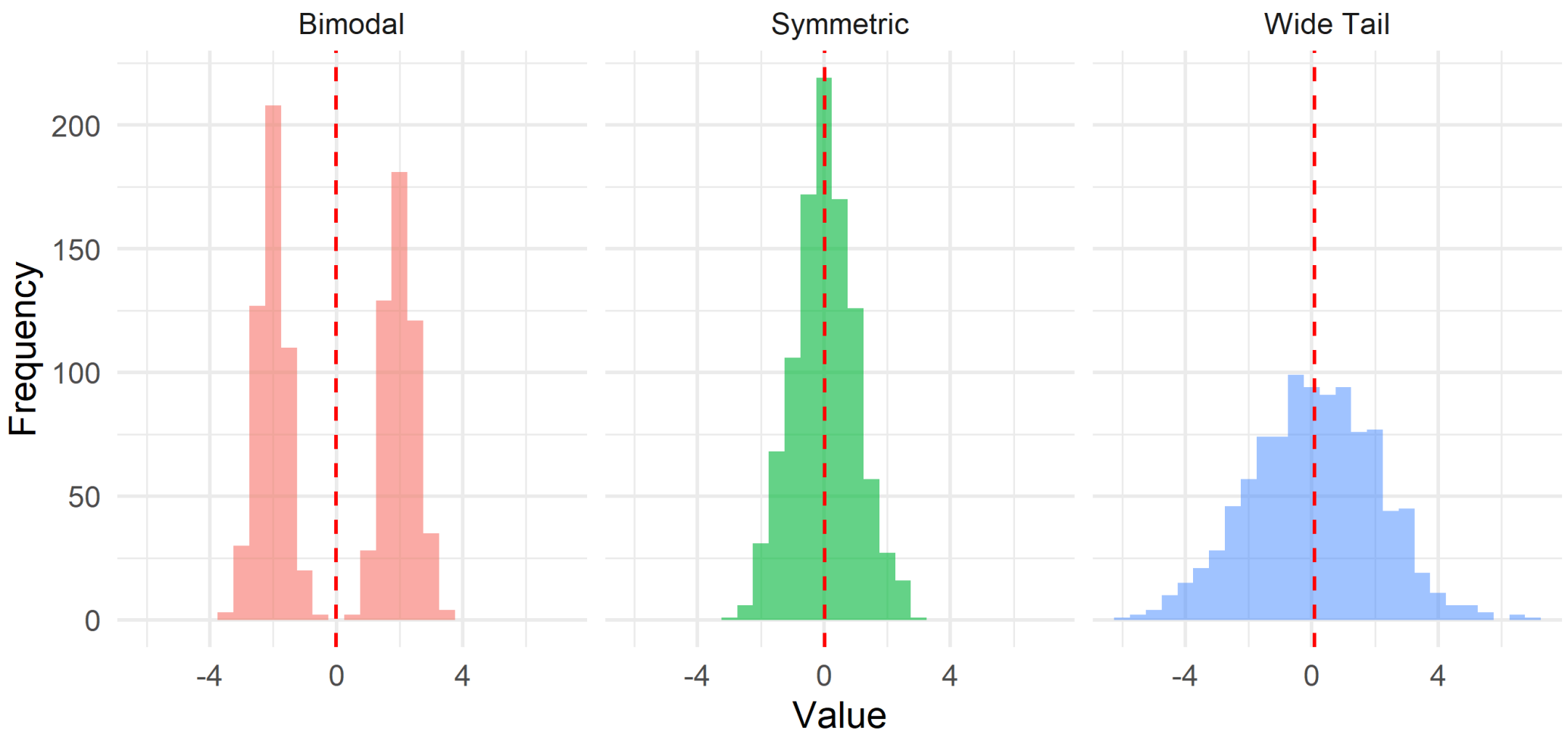
Consider a random variable X with $E(X) = \mu_x$ and standard deviation σ_x.

Ex: X is the number of followers a video gets, stored in `views`, distributed like this:



Same center — but rank them by spread

All three have a mean of ~0 (dashed line). Rank them from least to most spread out.



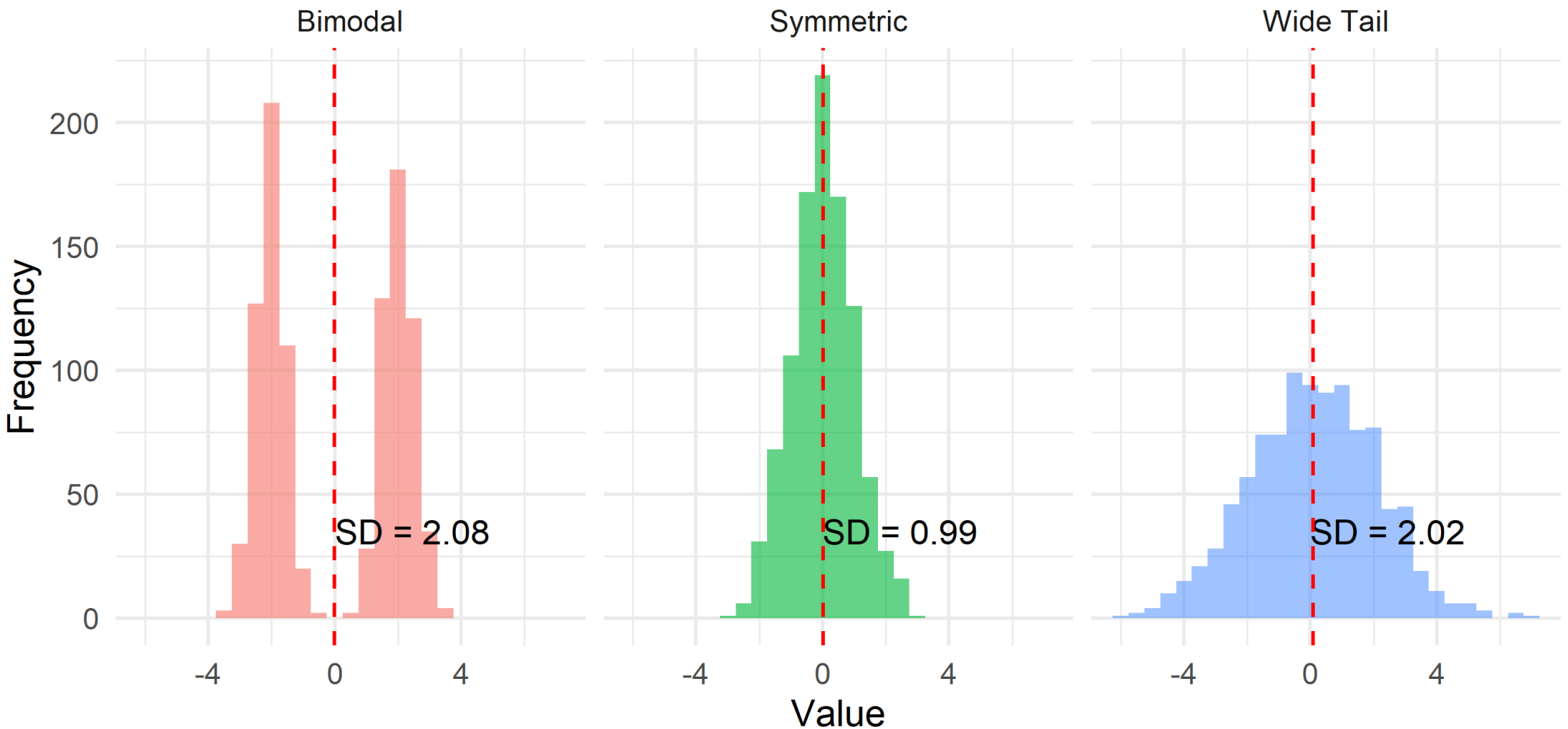
Exercise 15 — Rank them, then compute

The three samples from the previous slide are rebuilt for you as the vectors `symmetric`, `wide_tail` and `bimodal`. Commit to a ranking first, then compute the variance and the standard deviation of each and see whether your eye was right.

Exercise | Start Over | Show Hint | Run Code

```
1  set.seed(123)
2  symmetric <- rnorm(1000, mean = 0, sd = 1)
3  wide_tail <- rnorm(1000, mean = 0, sd = 2)
4  bimodal   <- c(rnorm(500, mean = -2, sd = 0.5), rnorm(500, mean = 2, sd = 0.5))
5
6  # my ranking, least to most spread: ____ , ____ , ____
```

Variance: same center, different spread



- Same centre in all three panels, three different standard deviations.
- The bimodal panel wins on spread without having a single extreme value — the SD measures distance from the centre, not the length of a tail.

Exercise 16 — Shift it, stretch it

The 20 follower counts in the picture are loaded as the vector `views`. Two questions — write your prediction first, then compute.

1. Everyone gains 40 followers: $Y = X + 40$. What happens to the mean of `views`? To the SD?
2. Everyone's followers triple: $Y = 3X$. What happens to the mean? To the SD? To the variance?

Exercise | Start Over | Show Hint | Run Code

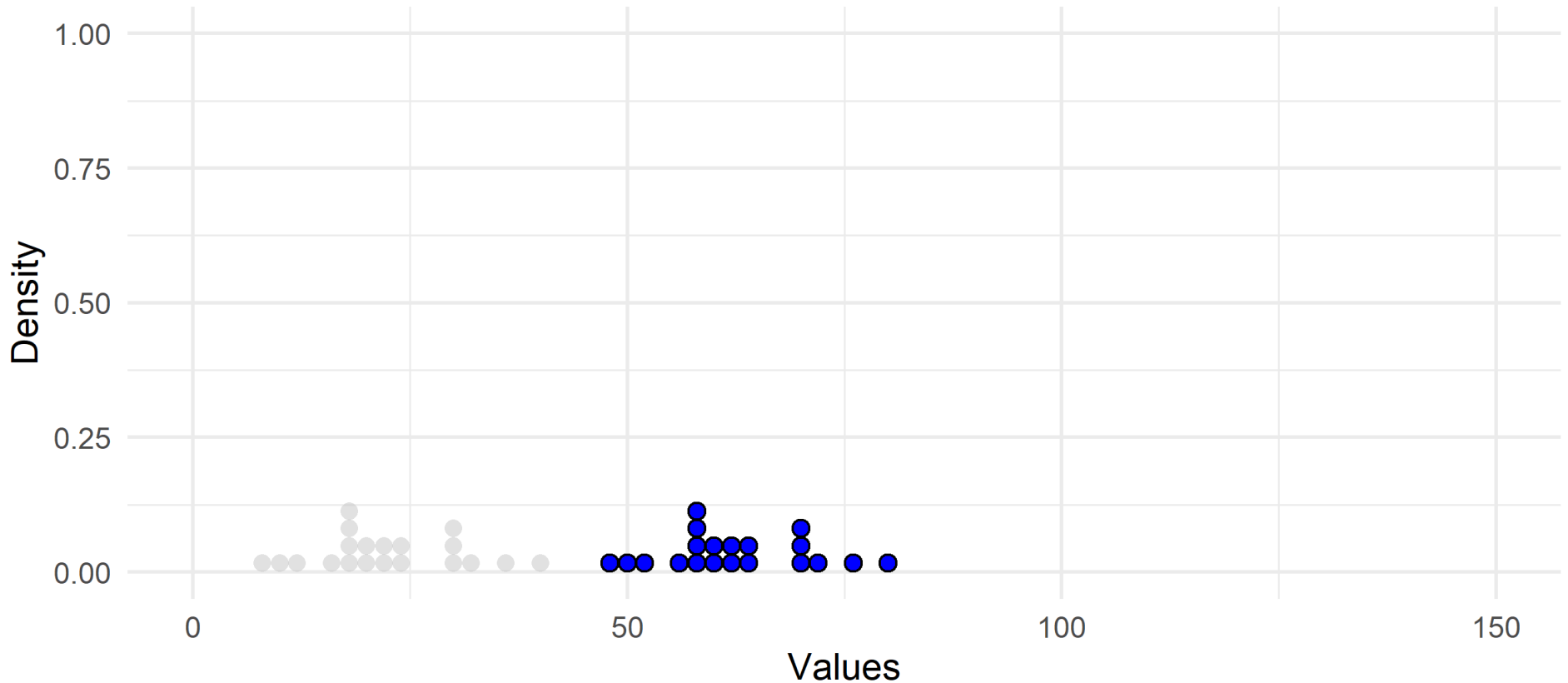
```
x <- views

mean(x)
sd(x)

# my prediction for x + 40 : mean ____ , sd ____
# my prediction for 3 * x  : mean ____ , sd ____
```

Standard Deviation: add a constant

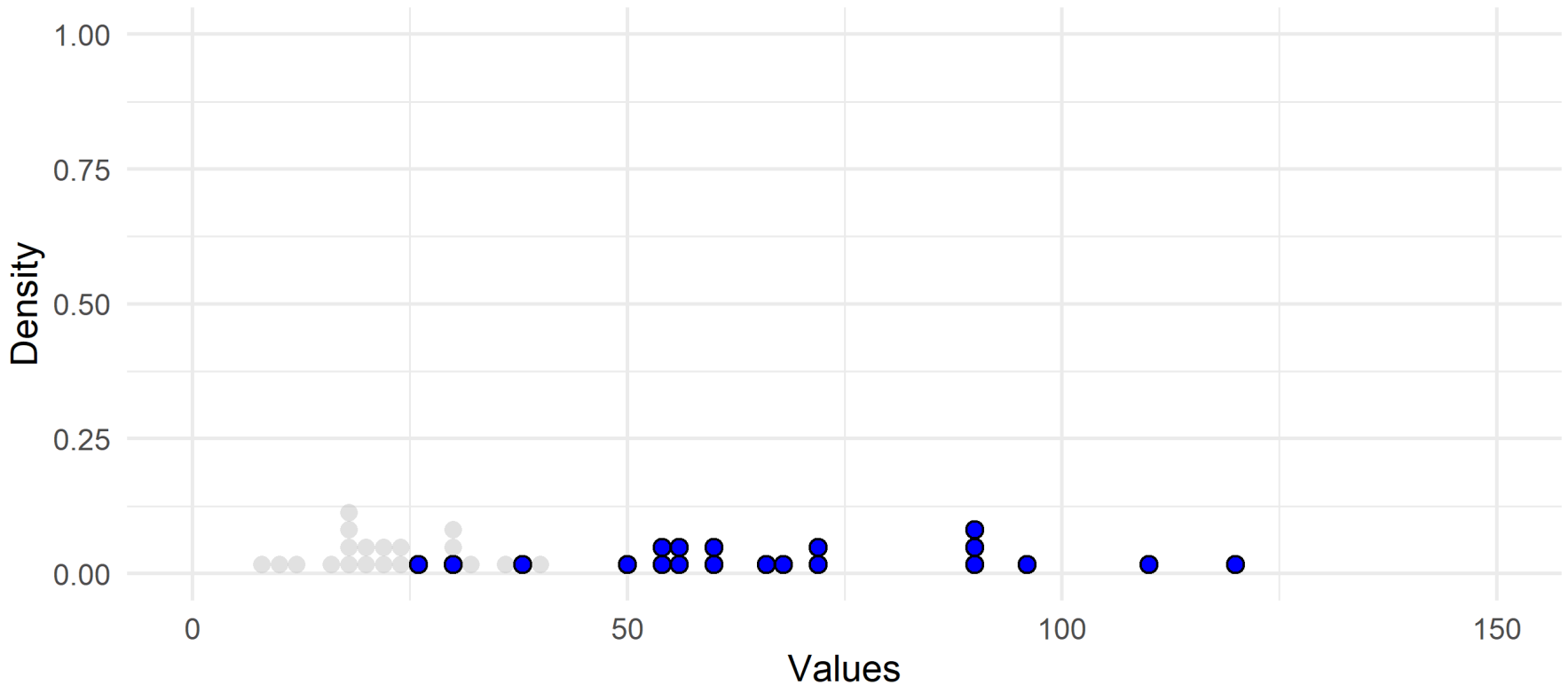
What happens to the mean and the standard deviation of `views` if everyone gets 40 more followers — $Y = X + 40$?



- Grey = the original counts, blue = after the shift.
- The whole cloud slides 40 to the right: the mean moves by 40, the distances between the points are untouched, so the SD does not move.

Standard Deviation: multiply by a constant

What happens to the mean and the standard deviation of `views` if everyone's followers get multiplied by 3 (without addition) — $Y = 3X$?



- Grey = the original counts, blue = after the stretch.
- The cloud is stretched, not shifted: every gap between two points is three times wider, so the SD is multiplied by 3 — and the variance by 9.

What a linear change does

- $Y = X + a$: the mean shifts by a, the SD does not move at all — every observation and the mean move together, so no distance changes.
- $Y = bX$: the mean is multiplied by b, the SD by $|b|$, and the variance by b^2.

$$E(X + c) = E(X) + c \qquad Var(X + c) = Var(X)$$

$$E(cX) = cE(X) \qquad Var(cX) = c^2 Var(X)$$

The algebra behind these four lines is in the appendix.

Coefficient of Variation

Coefficient of Variation divides the standard deviation by the mean.

$$C.V. = \frac{\sigma}{|\mu|}$$

And sample equivalent

$$c.v. = \frac{s}{|\bar{x}|}$$

- Why?
  - It expresses standard deviation as proportion of the mean
    - Small value means variation is low compared to the mean
  - It is unit free
  - You can compare it across variables with different units/magnitudes

Coefficient of Variation: example

Example — two stocks, priced in different currencies. Which one is riskier?

Date	MXN stock	USD stock
2023-07-01	91.59	1.01
2023-07-02	96.55	1.16
2023-07-03	123.38	1.02
2023-07-04	101.06	1.07
2023-07-05	101.94	1.09
2023-07-06	125.73	0.90
2023-07-07	106.91	1.35
2023-07-08	81.02	1.23

- One trades around 100 pesos, the other around 1.2 dollars.
- Their standard deviations are *not* comparable — different units, different magnitudes.
- So which stock moves around more?

🖊️ Exercise 17 — Make two incomparable things comparable

The two price series are rebuilt for you as the vectors `mxn` and `usd`. Compute the SD of each, then the coefficient of variation of each, and say which stock is more variable — and why the two statistics disagree about the answer.

Exercise | Start Over | Show Hint | Run Code

```
1 set.seed(123)
2 mxn <- round(rnorm(20, mean = 100, sd = 15), 2)
3 usd <- round(rnorm(20, mean = 1.2, sd = 0.18), 2)
```

Coefficient of Variation: the answer

- **Standard deviation**:
  - USD: 0.149
  - MXN: 14.59
- **Coefficient of variation**:
  - USD: 0.12
  - MXN: 0.14

The SD ratio is an artefact of the currency. The CV is unit-free, so it is the one you can compare.

Coefficient of Variation: transformations

So more generally, if $y_i = bx_i$, then

$$C.V._y = \frac{\sigma_y}{|\mu_y|} = \frac{|b|\sigma_x}{|b\mu_x|} = C.V._x$$

What if $y_i = bx_i + a$?

Then

$$C.V._y = \frac{\sigma_y}{|\mu_y|} = \frac{|b|\sigma_x}{|b\mu_x + a|} \neq C.V._x$$

Part 7 — The boxplot

Five numbers, one picture — and a rule that decides which points get called outliers.

Box and Whiskers plot

- Helps to see the distribution of the data and the outliers
  - Outliers reveal anomalies and potential data-collection errors
  - A whisker extends at most 1.5 × IQR from the box
  - Any point beyond that is flagged as an outlier
  - If no point is beyond 1.5 × IQR, the whisker stops at the last datapoint

✏️ Exercise 18 — Build one, then read the outliers off it

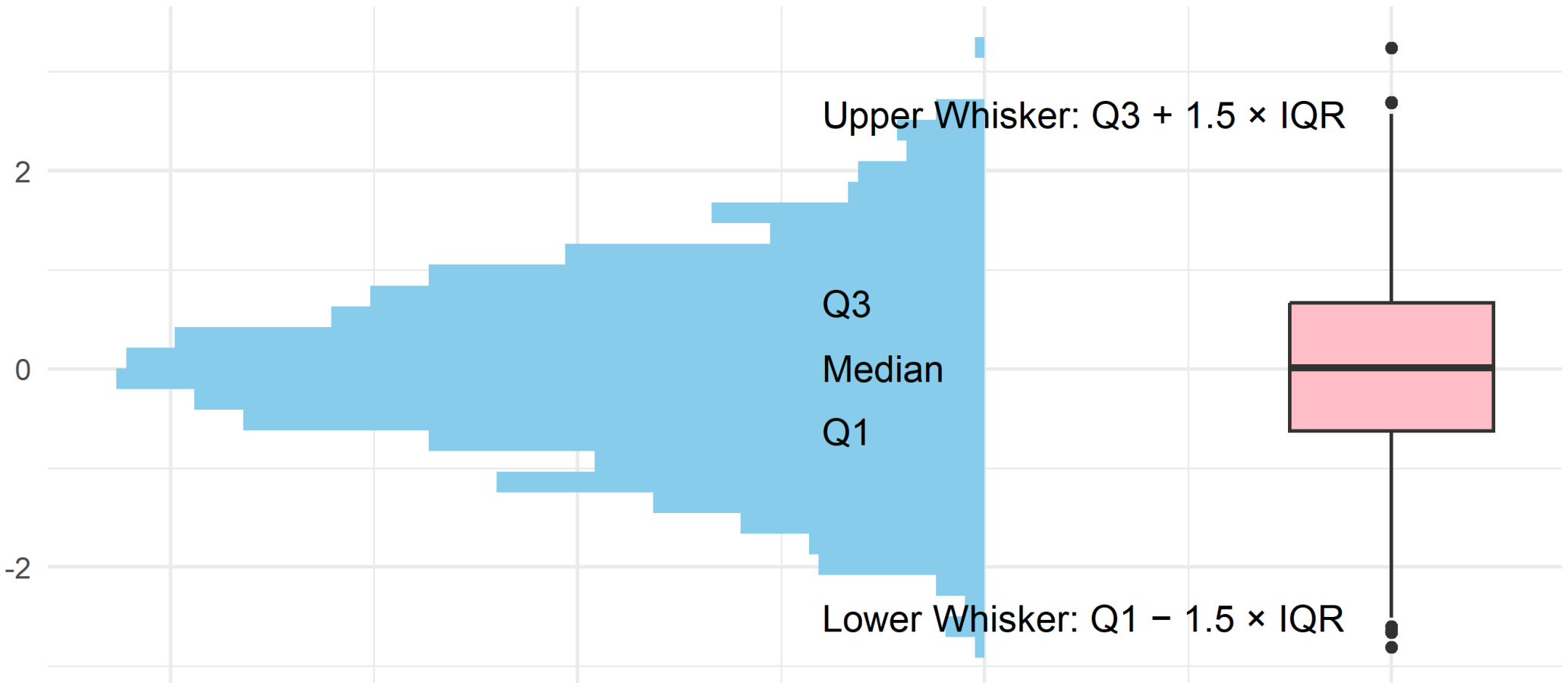
Draw a boxplot of `Health_data$weight` and print its quartiles. Then answer with numbers, not by eye: above which weight does R start drawing dots, and how many of the 37,858 people are out there?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
2
3 # Check your cut-off against the rule: q3 + 1.5 * (q3 - q1)
```

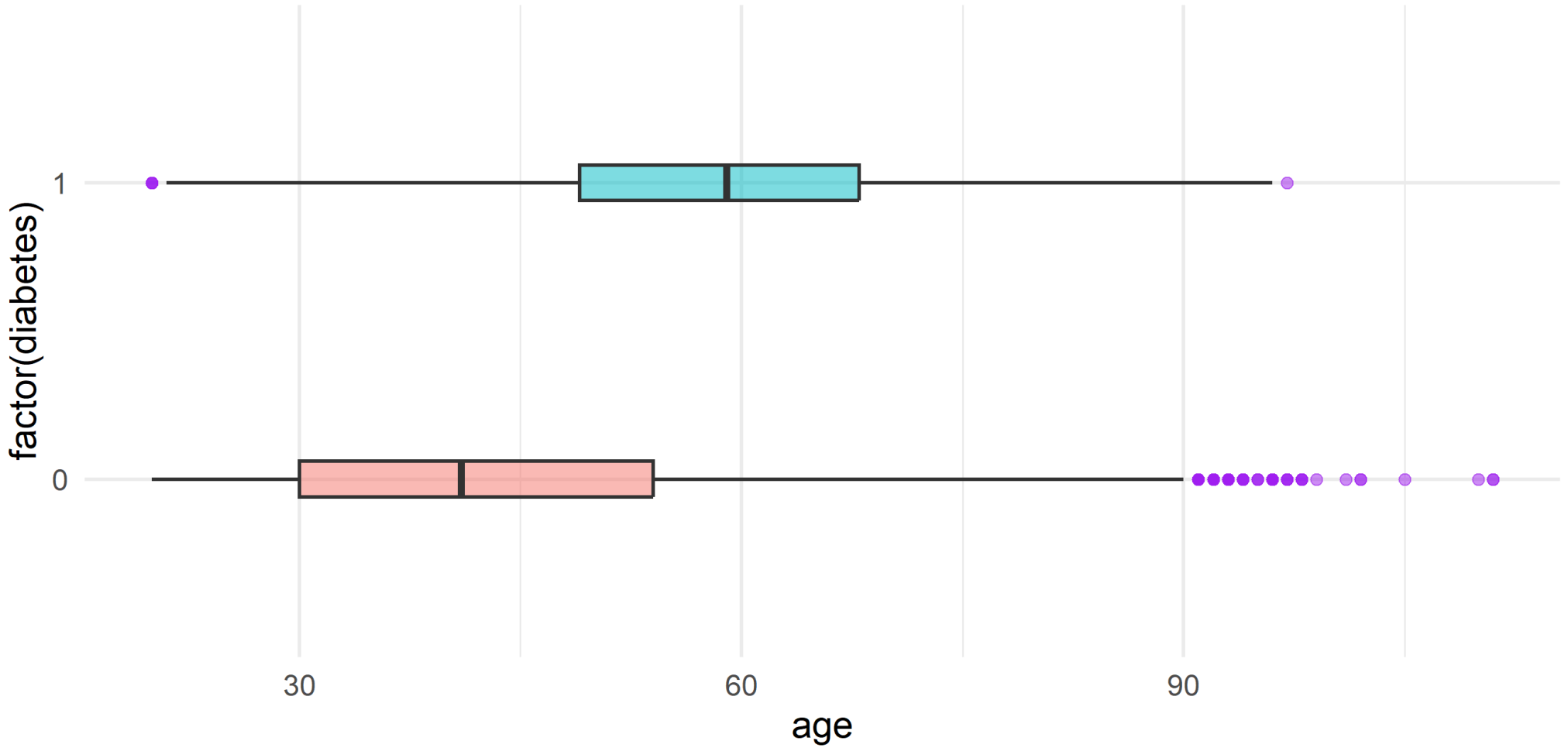
Box and Whiskers plot: anatomy

The box spans Q1–Q3 (the middle 50%); the line inside is the median.



Box and Whiskers plot (cont.)

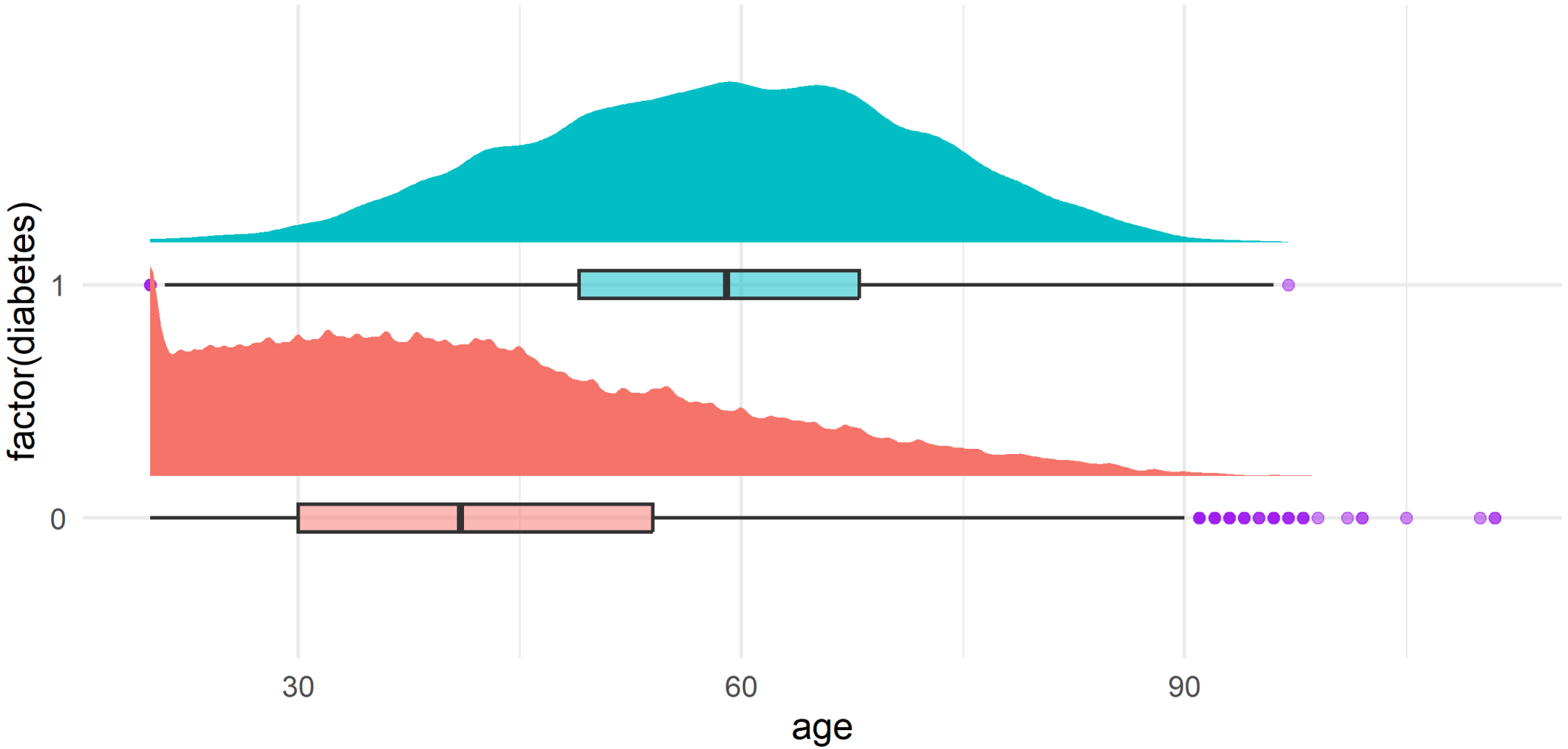
Dataset comparisons — they summarize data very well (age by diabetes status):



- Two groups, one picture: the median line answers “which group is higher”, the width of the box answers “which group is more spread out”.

Box and Whiskers plot (cont. 2)

Dataset comparisons — boxplot with the full distribution overlaid (age by diabetes status):

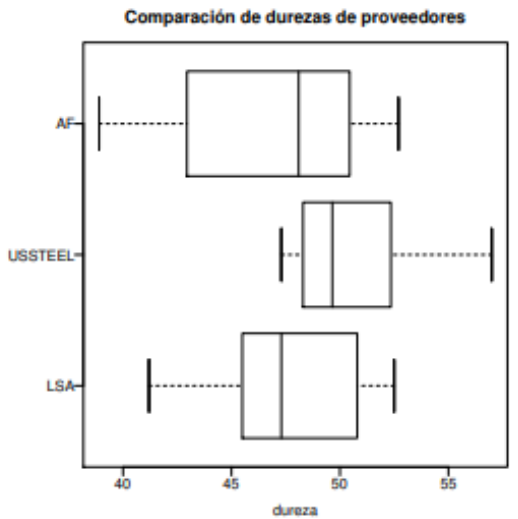


- The five numbers of the boxplot are a summary — the curve behind them shows what the summary threw away.

Reading a boxplot (warm-up)

5. [**5 puntos**] The next plot represents data on the hardness of steel rods for three different suppliers: AF, USSteel, and LSA. Based on this figure, it may be said that:

Hardness of steel rods for three different suppliers



a) Distributions for all supplier's hardness are skewed to the left.

b) AF's hardness seems to have the least dispersion of the three.

c) AF's hardness seems to have less dispersion than that of LSA.

d) USSTEEL hardness distribution is skewed to the right.

6. [**5 puntos**] The prior plot representing the hardness of steel rods for three different suppliers, allows us to identify useful:

a) central tendency and location measures.

b) insights about the distribution's dispersion.

c) asymmetry among distributions.

d) all of the above.

Exam spotlight — Fall 2025 Midterm 1

Real questions from last year’s Midterm 1. We solve these together in class.

→ More exam practice at the end of this deck

Exam question — Q1 & Q2

The Households & Environment Survey — the real Midterm 1 dataset — is loaded as `households`: 2,000 Mexican households surveyed once, in 2017.

Q1 (Dataset type). Is `households` cross-sectional, time-series, or panel?

Q2 (Variable classification). Classify each of `households$household_size`, `households$electricity_bill_monthly`, `households$separates_trash` and `households$locality_size`.

Exercise | Start Over | Show Hint | Run Code

```
1 # Find the evidence before you answer:
```

Exam question — Q3

`households` is loaded — 2,000 Mexican households from the 2017 Midterm 1 survey.

An energy company wants to understand the distribution of weekly household gasoline spending to design a fuel-discount program.

Visualize `households$gasoline_spending_weekly` and describe its shape.

Exercise | Start Over | Show Hint | Run Code

```
1 # Pick the right chart for this variable, then say what its shape is:
```

Exam question — Q5

`households` is loaded — 2,000 Mexican households from the 2017 Midterm 1 survey.

Compare `households$electricity_bill_monthly` between households where `households$uses_electricity_cooling` is `TRUE` and those where it is `FALSE`. Which group has the higher median bill?

Exercise | Start Over | Show Hint | Run Code

```
1 # Compare the two groups — a number and a picture:
```

Exercise 19 — The whole week, one question

`workforce` is loaded — 3,000 Mexican workers, with `workforce$monthly_income` and `workforce$gender` (`"Women"` / `"Men"`).

You are the compensation analyst. Your manager asks:

> *"Do women in this sample earn less than men — and by how much?"*

Answer it with the tools of this week: compare `workforce$monthly_income` across the two values of `workforce$gender`. Decide what to compute, compute it, and build one picture that supports your answer.

Exercise | Start Over | Show Hint | Run Code

```
1  # 1. A number: the typical income of each group. Which measure -- and why that one?
2
3  # 2. A picture: something a manager can read in three seconds.
4
5  # 3. One sentence: what do you tell your manager?
```

What you must be able to do this week

Data types & visualization

- Identify the unit of observation and whether data is cross-sectional, time-series, or panel.
- Classify a variable as categorical (nominal/ordinal) or numerical (discrete/continuous).
- Build and read the right chart for a categorical variable — a bar chart off a frequency table, where length = count and categories are easy to compare.

Central tendency

- Choose and compute the mean, median, and mode, and know which measure an outlier moves.
- Compute the mean of a binary variable (a proportion).
- Find and interpret percentiles, quartiles, and deciles.

First measures of spread

- Compute the range ($x_{max} - x_{min}$) and the IQR ($q_3 - q_1$) of a variable, and of separate groups within it.
- Explain why the range is destroyed by a single outlier and the IQR is not, and choose between them for a given question.

Spread that uses every observation

- Compute and interpret the variance and the standard deviation, and say why the sample version divides by $n - 1$.
- Know what a linear change $y = bx + a$ does to the mean, the SD and the variance.
- Compute the coefficient of variation and use it to compare spread across variables measured in different units.
- Choose the right spread measure for the question — robust (IQR) vs sensitive to outliers (variance, standard deviation).

The boxplot

- Build a boxplot and read its five-number summary — median, quartiles, whiskers — and say which points the $q_3 + 1.5 \times IQR$ rule flags as outliers.

Next week: the shape of the whole distribution — histograms, modality and skewness.

Practice for this week

Weekly exercise set: `W02_exercises.pdf`, with solutions in `W02_solutions.pdf`.

You can already solve exercises 1–13.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
12	Gender pay gap, ENOE salaries	R with the data
13	Household spending, ENIGH	R with the data
7	Store and loyalty card: shares within each store	R as a calculator
9	Median, quartiles and an outlier in complaints	R as a calculator
11	Covariance and correlation of advertising and sales	R as a calculator

Appendix — optional (extra & theory)

Variance & Standard Deviation — why square then root?

- Why do we first take squares and then take square root?
- Can't we just do $\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)$?
- NO! Because $\sum_{i=1}^{N}(x_i - \mu) = 0$

Variance & Standard Deviation (cont.) — why not MAD?

- Why don't we just do Mean Absolute Deviation?
  - $MAD = \frac{1}{N} \sum_{i=1}^{N} |(x_i - \mu)|$
- MAD is not differentiable at 0 :(
- Variance puts more weight on far away observations.
- It's a weighted average distance, where weights are distances themselves.

$$\underline{\sigma^2} = \sum_{i=1}^{N} (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^{N} \underbrace{|(x_i - \mu)|}_{Weight} * \underbrace{|(x_i - \mu)|}_{Distance}$$

- It makes it very sensitive to outliers, which get a lot of weight!
- Standard deviation retains this property

Standard Deviation (cont.) — $Y = X + 40$

$$\mu_Y = E(Y) = \frac{\sum_{i=1}^{N}(y_i + 40)}{N} = \frac{\sum_{i=1}^{N}(x_i + 40)}{N} = \frac{\sum_{i=1}^{N} x_i + N * 40}{N} = E(X) + 40 = \mu_X + 40$$

$$\sigma_Y = \sqrt{Var(Y)} = \sqrt{\frac{\sum_i (y_i - \mu_y)^2}{N}} = \sqrt{\frac{\sum_i ((x_i + 40) - (\mu_x + 40))^2}{N}} = \sqrt{\frac{\sum_i (x_i - \mu_x)^2}{N}} = \sqrt{Var(X)} = \sigma_X$$

Standard Deviation (cont.) — $Y = 3X$

$$\mu_Y = E(Y) = \frac{\sum_i y_i}{N} = \frac{\sum_i 3x_i}{N} = 3\frac{\sum_i x_i}{N} = 3E(X) = 3\mu_X$$

$$\sigma_Y = \sqrt{Var(Y)} = \sqrt{\frac{\sum_i (y_i - \mu_y)^2}{N}} = \sqrt{\frac{\sum_i (3x_i - 3\mu_x)^2}{N}} = \sqrt{3^2\frac{\sum_i (x_i - \mu_x)^2}{N}} = \sqrt{3^2 Var(X)} = 3\sigma_X$$

Exam practice

These are the Week 2 exercise-set problems, on real data: an excerpt of the household survey used in last year's Midterm 1, real daily transit ridership, ten years of Mexican labour-survey averages, a US labour-force survey, an inflation panel and 684 eBay auctions. Run the commands, look at what comes back, then decide.

← Back to the exam spotlight

Exam practice 1 — Data structures

Three real data frames are loaded: `households` (2,000 households, 2017 survey), `ridership` (daily city bus ridership) and `wages_panel` (Mexican labour-survey averages by region and year). Each is a different structure. Classify them — cross-sectional, time-series, or panel. [D1]

Don’t guess from the names. The answer is always the same two counts: how many units and how many time points per unit — so look for the unit column and the time column in each (`wages_panel$region` and `wages_panel$year`, `ridership$date`).

Exercise | Start Over | Show Hint | Run Code

```
# Look at each one first:
households[1:5, ]

# Then count units and time points. For example:
#   how many distinct years does wages_panel cover?
#   how many regions per year?
```

Exam practice 2 — Variable types

`households` is an excerpt of the actual Midterm 1 dataset — 2,000 Mexican households, surveyed once in 2017. Classify its columns as categorical, discrete, or continuous — and get the evidence out of R before you answer. [D2]

Start with `households$household_size`, `households$electricity_bill_monthly`, `households$separates_trash` and `households$locality_size` — the four the exam asked about.

Exercise | Start Over | Show Hint | Run Code

```
1  households[1:10, ]
2
3  # Then dig into single columns:
4  #   table(households$<col>)              few repeated labels or small whole numbers?
5  #   min / max / quantile                 for a column with hundreds of values
```

Exam practice 3 — Variable types II

Same three-way choice, from the description alone. [D3]

a. Daily number of Uber trips completed by one driver in Mexico City.

→ Discrete — a count of trips.

b. Whether an employee received job training during the past year.

→ Categorical — a yes/no label.

c. Time (in minutes) to serve a customer in a bank branch.

→ Continuous — a measurement; any positive value.

d. Number of times an individual has changed jobs in their career.

→ Discrete — a count.

📝 Exam practice 4 — Does the sampling strategy lie?

Monterrey wants the share of restaurants with health-inspection violations. Inspecting all of them is too costly, so they must sample. For each strategy, is there sample selection bias? [D4]

a. Every restaurant in one neighborhood. (b) A random 20% of the city.

b. All restaurants whose street number ends in 7. (d) The 40 largest by seating.

→ (a) bias — one neighborhood is not the city. (b) no bias — random sampling, the best strategy. (c) no bias — the last digit is unrelated to violations, so it is effectively random. (d) bias — the largest restaurants differ from the rest.

Exam practice 4 (cont.) — See it on real data

Same four strategies, but now you can check them against the truth: `households` (all 2,000 rows of the 2017 survey) is your whole "city", so you know the real answer.

Estimate the mean of `households$electricity_bill_monthly` under each strategy and compare it to the mean over all 2,000 rows. The strategies subset on `households$locality_size`, `households$household_id` and `households$household_size`. For (b), the stand-in for a random draw is every fifth `households$household_id` — the id number has nothing to do with `households$electricity_bill_monthly`, so selecting on it is as good as random.

Exercise | Start Over | Show Hint | Run Code

```
truth <- mean(households$electricity_bill_monthly, na.rm = TRUE)
truth

# (a) only households$locality_size == "Large_city"
# (b) every 5th households$household_id  (20% of the sample)
# (c) households$household_id ending in 7
# (d) the biggest households$household_size
```

Exam practice 5 — Selection bias in the wild

Is there a selection-bias concern, and why? [D5]

a. A pharma company recruits trial participants through newspaper ads and online forums.

→ Bias. People who answer ads differ systematically from the target population.

b. A university surveys only students who visit the career center.

→ Bias. Career-center visitors are more job-focused than the average student.

c. A supermarket chain samples receipts at random across all its stores.

→ No bias for its own customers — but it says nothing about people who never shop there.

d. A restaurant judges its food by its online reviews.

→ Bias. Reviews are self-selected: very happy and very angry customers write them.

e. A consulting firm surveys firms about profits; a third refuse to answer.

→ Bias (non-response). Firms that answer are likely the more successful ones.

Exam practice 6 — Exploring the cps dataset

The `cps` dataset (loaded for you) is a labor force survey with individuals' demographic characteristics, employment status, wages, and education. Variables of interest: [D6]

- `lfstatus`: labor force status (employed, unemployed, not in labor force)
- `marstatus`: marital status (married, single, divorced, widowed)
- `race`: race category (e.g., White, Black, Other)
- `wagehr`: hourly wage (in dollars, only observed for individuals paid by the hour)
- `ownchild`: number of own children in the household

a. Provide a frequency table of `cps$lfstatus`. (b) Draw a bar chart of the proportions of `cps$lfstatus`. Interpret the results.

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 6 — cps dataset (cont.)

c. Create a relative conditional frequency table of `cps$marstatus` and `cps$race`. (d) What is the difference in the probability of being married between Black and White individuals? [D6]

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 6 — cps dataset (cont. 2)

e. Calculate the difference in the average hourly wage (`cps$wagehr`) between married and not-married individuals. (f) Restrict to individuals aged 30–59 (`cps$age`): what are the mean and standard deviation of `cps$ownchild`? [D6]

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 6 — cps dataset (cont. 3)

g. The variable `cps$wagehr` has the following summary statistics: [D6]

```
   Min. 1st Qu. Median  Mean 3rd Qu.  Max.  NA's
   1.01   12.78  16.41  18.60   22.00 90.00  2174
```

i. Is the distribution symmetric, left-skewed, or right-skewed?

ii. If one more worker with wage = 16.41 is added, how would the sample mean and median change?

Get the evidence out of R before you answer — then check whether the table alone would have been enough to reach the same conclusion.

Exercise | Start Over | Show Hint | Run Code

```
1  # i.  compare the mean of cps$wagehr with its median
2
3  # ii. the extra worker is paid 16.41 -- which side of the mean is that?
```

i. Right-skewed. The mean (18.60) exceeds the median (16.41), and the distance from the median to the max (73.59) is much larger than from the min to the median (15.40). This is typical of wage distributions.

ii. The new wage equals the median, so the median stays the same or barely changes. The mean decreases slightly because 16.41 < 18.60 (the current mean), pulling the mean down.

Exam practice 6 — cps dataset (cont. 4)

(g), continued. Now compute: iii. the sample interquartile range (IQR) of `cps$wagehr`; iv. the values of the lower and upper whiskers in the box plot of `cps$wagehr`. [D6]

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 7 — Aguas frescas (by hand)

A small business sells aguas frescas at a weekend tianguis in Mexico City. Each cup is sold for 40 pesos. Over 16 Sundays, the number of cups sold per day has a sample mean of $\bar{x} = 120$ and a sample standard deviation of $s_x = 20$. [D8]

a. What are the sample mean and standard deviation of revenues in pesos? (*Revenue = price × quantity.*)

b. The stall rental fee is 2,000 pesos per Sunday, and it costs 5 pesos to produce each cup. What are the sample mean and standard deviation of profits? (*Profit = Revenue − Fixed Cost − Variable Cost.*)

Use R as a calculator — apply the linear-transformation rules for mean and SD:

Exercise | Start Over | Show Hint | Run Code

```
1 xbar <- 120
2 s_x  <- 20
3 # Your code here
```

Exam practice 8 — Inflation panel

Use the `inflation` dataset (loaded for you): annual inflation rates for 45 countries, 2010–2019. Variables: `inflation$country` (3-letter code), `inflation$year`, `inflation$inflation` (percentage points). Briefly interpret each result. [D9]

a. How many observations are associated with deflation (negative inflation rates)?

b. How many countries experienced deflation at least once during this period?

c. Which countries have the lowest and highest *average inflation* over the 10-year period?

d. Which countries have the lowest and highest *standard deviation of inflation*?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 8 — Inflation panel (cont.)

e. Draw a time-series plot for inflation in the United States (`inflation$country == "USA"`). (f) Draw a time-series plot showing inflation for USA, Canada (CAN), and Mexico (MEX) on the same plot — make sure each country's line is distinguishable. [D9]

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 9 — Wages by race

Use the `cps` dataset again. [D10]

a. Plot the average hourly wage (`cps$wagehr`) by race (`cps$race`). What does it show? Report the group means and compare across races. Interpret carefully: are there differences in wages across demographic groups? How might economists explain them?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 10 — eBay auctions

Use the `auctions` dataset (loaded for you): 684 eBay auctions of iPod Mini devices in 2006. Key variables: `auctions$finalprice` (winning bid), `auctions$bidders` (number of unique bidders), and condition dummies (`auctions$new`, `auctions$used`, `auctions$refurb`). [D11]

a. Focus only on the subsample of 624 new and used devices (drop those with `auctions$refurb` = 1).

b. Draw a scatter plot of `finalprice` versus `bidders`.

c. Compute the sample correlation between `finalprice` and `bidders`. Why is this value interesting in the context of auctions?

d. Based on the plot in (b), does `finalprice` always increase with more bidders, or is the relationship more nuanced? Explain.

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exam practice 11 — Units of measurement

For a sample of Mexican manufacturing firms we observe $x =$ electricity purchased (in kilowatt-hours, kWh) and $y =$ monthly revenues (in thousands of pesos). [D12]

Build such a sample and watch what each statistic does when you change the units of y from thousands of pesos to pesos. The statistic whose value does not move is the one with no units.

Exercise | Start Over | Show Hint | Run Code

```r
set.seed(1)
x <- round(rlnorm(200, 9, 0.4))                # electricity, kWh
y <- round(0.004 * x + rnorm(200, 0, 5), 1)    # revenue, thousands of pesos

# mean(x), var(y), cov(x, y), cor(x, y) -- then redo with y in pesos (y * 1000)
```

Now answer in words. What are the units of:

a. the sample average of x?

kWh — the mean has the same units as the variable.

b. the variance of y?

(thousands of pesos)^2 — variance squares the deviations, so it squares the units.

c. the covariance of x and y?

kWh × thousands of pesos — the product of the two variables' units.

d. the correlation of x and y?

Unitless — the units cancel in the ratio $s_{xy} / (s_x \cdot s_y)$.

Exam practice 11 — Units of measurement (cont.)

Suppose the sample summary statistics for salary (y, in thousands of pesos) and education (x, in number of years) are: $\bar{x} = 200$, $s_x = 30$, $\bar{y} = 17$, $s_y = 3$, and $s_{xy} = 50$. [D12]

a. What is the sample correlation between salary and education?

b. What would the sample covariance be if salary were instead expressed in pesos (not thousands)?

Exercise | Start Over | Show Hint | Run Code

```
1 s_x  <- 30
2 s_y  <- 3
3 s_xy <- 50
4 # Your code here
```

Exam practice 12 — Portfolio variance

Consider monthly returns of two Mexican stocks, A and B, listed on the *Bolsa Mexicana de Valores* (BMV). Suppose $\mathrm{Var}(x) = 0.006$ and $\mathrm{Var}(y) = 0.008$. [D13]

a. What condition must the correlation $\mathrm{Corr}(x, y)$ satisfy for the variance of the portfolio return $\frac{1}{2}(x + y)$ to be less than or equal to 0.007?

Work it out numerically — find the covariance bound, translate it into a correlation bound, and check the worst case $\rho = 1$:

Exercise | Start Over | Show Hint | Run Code

```
1 var_x <- 0.006
2 var_y <- 0.008
3 # Your code here
```

Exam practice 13 — Hours, wages & education

Use the `cps` dataset. Key variables for this exercise: `cps$hrslastwk` (hours worked last week), `cps$wagehr` (hourly wage in US dollars), `cps$educ` (years of education), `cps$gender` ("Male" or "Female"). [D14]

a. Calculate the average number of hours worked last week (`cps$hrslastwk`) by gender. What differences do you observe? How might labor market factors (e.g., childcare responsibilities, occupational segregation) explain these patterns?

b. Calculate the covariance between wages (`cps$wagehr`) and education (`cps$educ`). Does its sign suggest a positive or negative relationship?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

📝 Exam practice 13 — Currency conversion (cont.)

c. Suppose wages (`cps$wagehr`) are converted into Mexican pesos (assume 1 USD = 20 MXN). [D14]

i. What happens to the covariance between wages and education when wages are rescaled?

ii. What happens to the correlation?

Verify both numerically by rescaling `cps$wagehr` yourself, then interpret: why does correlation remain unchanged even though covariance changes?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```